

The Trace Formula for the Impatient

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1 Tools

1.1 The Adelic Quotient

Notation 1. Throughout, F is a global field, and $V(F)$ is the set of places of F , and $v \in V(F)$ is a place.

Notation 2. Throughout, G is an affine algebraic group, usually reductive.

Notation 3. If F is a number field, define A_G to be the identity component of $A(\mathbb{R})$, where $A \subseteq \text{Res}_{F/\mathbb{Q}} Z_G$ is the largest split torus. If F is a function field of characteristic p , let Z be the largest split torus in $\text{Res}_{F/\mathbb{F}_p(t)} Z_G$, and define $A_G \subseteq Z(\mathbb{F}_p(t)_\infty)$ to be the inverse image of the subgroup $(t^\mathbb{Z})^d$ under a choice of isomorphism $Z \cong \mathbb{G}_m^d$.

Notation 4. Set

$$G(\mathbb{A}_F)^1 := \bigcap_{\chi \in X^*(G)} \ker(|\cdot| \circ \chi).$$

Notation 5. Set $[G] := A_G G(F) \backslash G(\mathbb{A}_F)$.

Proposition 6. If F is a number field, the natural map $G(F) \backslash G(\mathbb{A}_F)^1 \rightarrow [G]$ is a homeomorphism. If F is a function field, then $A_G G(\mathbb{A}_F)^1$ has finite index in $G(\mathbb{A}_F)$.

Theorem 7. The quotient $G(F) \backslash G(\mathbb{A}_F)^1$ has finite volume. It is compact if and only if every F -split torus $T \subseteq G$ has $T_{\overline{F}} \subseteq R(G_{\overline{F}})$.

Theorem 8 (Dixmier–Malliavin). Suppose G is reductive. Then any $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$ is a finite sum of convolutions of pairs of functions in $C_c^\infty(A_G \backslash G(\mathbb{A}_F))$.

1.2 Decompositions

Theorem 9 (Levi). Suppose G is reductive, and let $P \subseteq G$ be a parabolic subgroup. Let $N \subseteq P$ be the unipotent radical. Then the short exact sequence

$$1 \rightarrow N \rightarrow P \rightarrow P/N \rightarrow 1$$

splits.

Theorem 10 (Iwasawa). Suppose G is reductive, and let $P \subseteq G$ be a parabolic subgroup. Then there is a maximal compact subgroup $K \subseteq G(\mathbb{A}_F)$ for which

$$G(\mathbb{A}_F) = P(\mathbb{A}_F)K.$$

Notation 11. Suppose G is reductive, and let $P \subseteq G$ be a parabolic subgroup with Levi decomposition $P = MN$. Define $\mathfrak{a}_M^* := X^*(M)_{\mathbb{R}}$ and $\mathfrak{a}_M := \text{Hom}(X^*(M), \mathbb{R})$. Then define $H_M: M(\mathbb{A}_F) \rightarrow \mathfrak{a}_M$ via

$$\langle H_M(m), \lambda \rangle := \log |\lambda(m)|.$$

Because $K \cap M(\mathbb{A}_F) \subseteq \ker H_M$, the function $H_P: G(\mathbb{A}_F) \rightarrow \mathfrak{a}_M$ given by $H_P(nmk) := H_M(m)$ is well-defined.

Remark 12. Suppose F is a number field. Then the restriction $H_P: A_M \rightarrow \mathfrak{a}_M$ is an isomorphism: it is an iterated product of the isomorphism $\log: \mathbb{R}^+ \rightarrow \mathbb{R}$.

Definition 13 (Siegel set). Suppose G is reductive, and let $P \subseteq G$ be a minimal parabolic subgroup with Levi decomposition $P = MN$ and Iwasawa decomposition $G(\mathbb{A}_F) = P(\mathbb{A}_F)K$. Let $Z_0 \subseteq Z_G$ and $T_0 \subseteq G^{\text{der}}$ be maximal split tori. Then a *Siegel set* in $G(\mathbb{A}_F)^1$ is a set of the form

$$\mathfrak{S}(t) := \Omega_N \Omega_M A_{T_0}(t) K,$$

where $\Omega_N \subseteq N(\mathbb{A}_F)$ and $\Omega_M \subseteq M(\mathbb{A}_F)^1$ are compact, and

$$A_{T_0}(t) := \{x \in A_{T_0} : |\alpha(x)|_{\infty} \geq t \text{ for all } \alpha \in \Delta\}.$$

Theorem 14 (Reduction theory). Suppose G is reductive. Then there is a Siegel set $\mathfrak{S}(t)$ such that

$$G(F)\mathfrak{S}(t) = G(\mathbb{A}_F)^1.$$

Remark 15. For $x \in \mathfrak{S}(t)$, the $s_x \in A_{T_0}(t)$ piece is actually independent of the choice decomposition. Indeed, simply invert $H_P: \mathfrak{S}(t) \rightarrow \mathfrak{a}_M$.

Theorem 16 (Cartan). Let $P_0 \subseteq G$ be a minimal parabolic subgroup with maximal split torus $A_0 \subseteq P_0$, and let $K_0 \subseteq G(F_v)$ be a maximal compact subgroup from the Iwasawa decomposition. If v is finite, then

$$G(F_v) = \bigcup_{\lambda \in X_*(A_0)^+} K_0 \varpi^\lambda K_0.$$

Sketch. In the split case, define $K_0 := G(\mathcal{O}_v)$. ■

Theorem 17 (Iwahori). Suppose v is finite and G is reductive. Let $P_0 \subseteq G$ be a minimal parabolic subgroup, and let $K_0 \subseteq G(F_v)$ be a maximal compact open subgroup with Iwasawa decomposition $G(F_v) = P_0(F_v)K_0$. Let $P_0 = MN^+$ be the Levi decomposition, and let N^- consist of the negative root spaces. Then there is compact open neighborhood basis $\{K_n\}_{n \geq 0}$ of $1 \in G(F_v)$ such that the multiplication maps

$$(N^+ \cap K_n) \times (M \cap K_n) \times (N^- \cap K_n) \rightarrow K_n$$

is an isomorphism for all n .

Sketch. In the split case, define K_n to be the kernel of $G(\mathcal{O}_v) \rightarrow G(\mathcal{O}_v/\mathfrak{p}^n)$. In general, pass to the split case by a Galois descent argument. ■

1.3 Functional Analysis

Definition 18. Let V be a Hilbert space. For a bounded operator $A: V \rightarrow V$, define the *adjoint* $A^*: V \rightarrow V$ to be the unique linear function such that

$$\langle Av, w \rangle_V = \langle v, A^*w \rangle_V$$

for all $v, w \in V$. It turns out that A^* is a bounded operator.

Definition 19. Let V be a Hilbert space. An operator $A: V \rightarrow V$ is *positive* if and only if $\langle Av, v \rangle_V \geq 0$ for all $v \in V$.

Lemma 20. Let A be a positive bounded operator on a Hilbert space V . Then A admits a unique positive square $\sqrt{A}$ for which $\sqrt{A} \circ \sqrt{A} = A$.

Notation 21. Let A be a bounded operator on a Hilbert space V . Then we define

$$|A| := \sqrt{A \circ A^*}.$$

Definition 22 (compact). A bounded linear operator $A: V \rightarrow V$ is *Hilbert–Schmidt* if and only if there is an orthonormal basis $\{v_i\}_i$ of V such that

$$\sum_i \|Av_i\|_V^2 < \infty.$$

If A is Hilbert–Schmidt, we define $\|A\|_{\text{HS}} := \left(\sum_i \|Av_i\|_V^2\right)^{1/2}$.

Definition 23 (trace class). A bounded linear operator $A: V \rightarrow V$ is *trace class* if and only if there is an orthonormal basis $\{v_i\}_i$ of V such that

$$\sum_i (|A| v_i, v_i) < \infty.$$

If A has trace class, we define $\|A\|_{\text{tr}} := \sum_i (|A| v_i, v_i)$ and $\text{tr } A := \sum_i (Av_i, v_i)$.

Remark 24. The norms and the trace are independent of the choice of basis, but this is tricky to show.

Example 25. Let X and Y be compact Hausdorff spaces equipped with Radon measures. Choose $K \in C(X \times Y) \cap L^2(X \times Y)$. Then the integral operator $T: L^2(X) \rightarrow L^2(Y)$ defined by

$$Tf(y) := \int_X K(x, y) f(x) dx$$

is bounded and Hilbert–Schmidt. One sees this by comparing T to certain “Riemann sum operators,” which are finite rank.

Remark 26. Trace class implies Hilbert–Schmidt.

Theorem 27 (Spectral). Let $A: V \rightarrow V$ be a self-adjoint Hilbert–Schmidt operator on a Hilbert space V . Then there is a decomposition

$$V = \ker A \oplus \bigoplus_{\lambda \in \text{Spec } A} V[\lambda],$$

where $\text{Spec } A \subseteq \mathbb{R}$ is a discrete set, except possibly with a limit point at zero, and $V[\lambda] := \ker(A - \lambda 1_V)$ is finite-dimensional.

Sketch. Inductively strip one eigenvalue off at a time. ■

Corollary 28. Let (π, V) be a unitary representation of a locally compact Hausdorff group G . Suppose that $\{f_n\}$ is an approximate identity in $C_c(G)$ such that $\pi(f_n)$ is Hilbert–Schmidt for all n . Then V is a Hilbert direct sum of its irreducible subrepresentations, each of which has finite multiplicity.

Sketch. This is fairly technical. To show that the multiplicities are finite, for any irreducible π' , choose $f \in C_c(G)$ with $\pi'(f) \neq 0$ so that there is a nonzero eigenvalue λ of $\pi'(f)$. Then choose some $v' \in \pi'$, and evaluation at v' shows

$$\dim \text{Hom}_G(\pi', \pi) \leq \dim \ker(f - \lambda; V),$$

which is finite by the Spectral theorem.

To exhibit the decomposition, we use Zorn’s lemma. By a Zorn’s lemma argument with orthogonal complements, it is enough to show that any nonzero subrepresentation $W \subseteq V$ admits an irreducible subrepresentation. We may as well assume $W = V$. Let’s describe how to construct the irreducible subrepresentation: choose $f \in C_c(G)$ with $\pi(f) \neq 0$. Then choose a nonzero eigenvalue λ of $\pi(f)$; note $V[\lambda]$ is finite-dimensional. Define M to be a nonzero subspace minimal among the intersections

$$\{V[\lambda] \cap V' : V' \subseteq V \text{ is a subrepresentation}\} \setminus \{\emptyset\}.$$

The subrepresentation generated by any nonzero vector in M will work! ■

Proposition 29. Let X be a locally compact second countable Hausdorff space equipped with a Radon measure. Let $\mathcal{HS}(X)$ denote the Hilbert space of Hilbert–Schmidt operators on $L^2(X)$. Then $\mathcal{HS}(X) \cong L^2(X \times X)$.

Sketch. The map $L^2(X \times X) \rightarrow \mathcal{HS}(X)$ is given by sending a kernel function to the corresponding integral operator. For the inverse map, fix a basis $\{\varphi_i\}_i$ of $L^2(X)$. Then send $A \in \mathcal{HS}(X)$ to the function

$$K_A(x, y) := \sum_i A\varphi_i(x) \overline{\varphi_i(y)}.$$

In particular, one can show $A \mapsto K_A$ is an isometry. ■

Remark 30. The proof shows that the function

$$\sum_i A\varphi_i(x) \overline{\varphi_i(x)}$$

does not depend on the choice of basis.

Proposition 31. Let X be a locally compact second countable Hausdorff space equipped with a Radon measure. Let $A: L^2(X) \rightarrow L^2(X)$ be an integral operator given by a continuous kernel $K \in L^2(X \times X)$. Then

$$\mathrm{tr} A = \int_X K(x, x) dx.$$

Sketch. This is actually pretty tricky. Intuitively, on a basis $\{\varphi_i\}$ of $L^2(X)$, we find that $\mathrm{tr} A$ equals

$$\sum_i \langle A\varphi_i, \varphi_i \rangle = \sum_i \int_X A\varphi_i(x) \overline{\varphi_i(x)} dx.$$

Rearranging the sum with the integral completes the proof. ■

Remark 32. In the application, we will have $X = [G]$ and A will usually be given by Rf . Then one can use the Dixmier–Malliavin theorem to reduce to the case where f is a convolution $f_1 * f_2$, so the trace is a product of Rf_1 and Rf_2 . In this case, convergence is easier to show.

2 Nonarchimedean Representation Theory

Throughout, v is a finite place.

2.1 The Hecke Algebra

Definition 33 (smooth). A representation V of $G(F_v)$ is *smooth* if and only if each $v \in V$ is stabilized by a compact open subgroup of $G(F_v)$.

Remark 34. Let (π, V) be a smooth finite-dimensional representation of $G(F_v)$, where G is split reductive. Then we claim that π factors through $G^{\mathrm{ab}}(F_v)$. This follows from the Iwahori decomposition: $\ker \pi$ contains an open subgroup, so it contains K_n for some $n > 0$, so it contains a compact open subgroup of each root space N_α . Because $\ker \pi$ is normal, we may conjugate this subgroup to cover N_α . Lastly, the root spaces generate G^{der} .

Definition 35 (admissible). A representation V of $G(F_v)$ is *admissible* if and only if it is smooth and

$$\dim V^K < \infty$$

for each compact open subgroup $K \subseteq G(F_v)$.

Definition 36 (Hecke algebra). Let $\mathcal{H}_G$ denote the *Hecke algebra*, which is the algebra of smooth distributions $\mu: C_c^\infty(G(F_v)) \rightarrow \mathbb{C}$. For any compact open $K \subseteq G(F_v)$, define $\mathcal{H}_{G,K}$ as those distributions which are K -invariant on the left and right. The theory of the Haar measure allows us to identify $\mathcal{H}_G$ with $C_c^\infty(G(F_v))$ (uniquely up to scalar), which we will frequently do.

Remark 37. For any smooth representation V of $G(F_v)$, note that $\mathcal{H}_G$ acts on V by

$$\mu \cdot f := \int_{G(F_v)} f(x) d\mu(x).$$

Notation 38. For each compact open subgroup $K \subseteq G(F_v)$, define $e_K \in \mathcal{H}_G$ as

$$\frac{1}{\text{vol}(K; dx)} \cdot 1_K dx \in \mathcal{H}_G,$$

where dx is a choice of Haar measure on $G(F_v)$.

Remark 39. If V is smooth, e_K acts by projection $V \rightarrow V^K$.

Lemma 40. Fix a smooth representation V of $G(F_v)$. Then V is irreducible if and only if V^K is zero or irreducible as a representation of $\mathcal{H}_{G,K}$ for each $K \subseteq G(F_v)$.

Sketch. In the reverse direction, note that a nonzero factorization of V remains nonzero after taking invariants for small enough K by smoothness. In the forward direction, note that if $W \subseteq V^K$ is a nonzero subrepresentation, then $\mathcal{H}_G \cdot W = V$, so $\mathcal{H}_{G,K} \cdot W = V^K$. ■

Lemma 41. Fix smooth irreducible representations V_1 and V_2 of $G(F_v)$. Choose $K \subseteq G(F_v)$ for which V_1^K and V_2^K are nonzero. If $V_1^K \cong V_2^K$, then $V_1 \cong V_2$.

Sketch. Choose an isomorphism $I: V_1^K \rightarrow V_2^K$, and fix nonzero $v \in V_1^K$. It is enough to check that the extension

$$I(f \cdot v_1) := f \cdot I(v_1)$$

is a well-defined map $V_1 \rightarrow V_2$. Thus, we want to show $f \cdot v_1 = 0$ implies $f \cdot I(v_1) = 0$. This is tricky: one can show $e_K f_0 f_1 \cdot I(v_1) = 0$ for any f_0 , so $\mathcal{H}_{G,K} \cdot fI(v_1) = 0$, but $\mathcal{H}_{G,K} \cdot fI(v_1) = (\mathcal{H}_G \cdot fI(v_1))^K$, so $\mathcal{H}_G \cdot fI(v_1) \subseteq V_2^K$, so $fI(v_1) = 0$ by irreducibility! ■

2.2 Parabolic Induction and Jacquet Modules

Definition 42 (contragredient). Let V be a smooth representation of $G(F)$. Then define $V^\vee$ to consist of the smooth linear functionals on V , equipped with the natural $G(F)$ -action.

Remark 43. If V is admissible, $V^\vee$ is admissible, and $V \cong V^{\vee\vee}$.

Definition 44 (parabolic induction). Let $P \subseteq G$ be a parabolic subgroup with Levi decomposition $P = MN$. For a representation (π, V) of $M(F_v)$, we define $\text{Ind}_P^G \pi$ to be the representation

$$\left\{ \varphi \in C^\infty(G(F_v), V) : \varphi(mng) = \delta_P^{1/2}(m) \cdot m\varphi(g) \right\},$$

where $G(F_v)$ acts by right translation.

Remark 45. If π is smooth, then $\text{Ind}_P^G \pi$ is smooth. If π is admissible, then $\text{Ind}_P^G \pi$ is admissible.

Remark 46. If π is unitary, then $\text{Ind}_P^G \pi$ is unitary. In short, choose a maximal compact subgroup $K \subseteq G(F_v)$ with $G(F_v) = P(F_v)K$. Then define the inner product

$$\langle \varphi, \psi \rangle_{\text{Ind}_P^G \pi} := \int_K \langle \varphi(x), \psi(x) \rangle dx.$$

(The same inner product can be used to show that $(\text{Ind}_P^G \pi)^\vee \cong \text{Ind}_P^G \pi^\vee$.) To see that this is invariant, use the Iwasawa decomposition and rewrite the integral as $\int_{G(F_v)} \langle \varphi(x), \psi(x) \rangle f(x) dx$, where $f \in C_c^\infty(G(F_v))$ satisfies $\int_{P(F_v)} f(pg) dp = 1$ for all g .

Definition 47 (Jacquet module). Let $P \subseteq G$ be a parabolic subgroup with Levi decomposition $P = MN$, and let $\psi: N(F_v) \rightarrow \mathbb{C}^\times$ be a character. For a representation V of $G(F_v)$, define the *Jacquet module* $J_{\psi, N}(V)$ as

$$V/V(N, \psi) \otimes \delta_P^{-1/2},$$

where $V(N, \psi) := \text{span}\{n\varphi - \psi(n)\varphi : \varphi \in V, n \in N(F_v)\}$. If $\psi = 1$, we may omit it from the notation. We may omit N from the notation. Note that $J_{\psi, N}(V)$ is a representation of $M(F)$.

Remark 48. If V is smooth, then $J(V)$ is smooth.

Proposition 49. Let $P \subseteq G$ be a parabolic subgroup with Levi decomposition $P = MN$. The functors $J: \text{Rep } G(F_v) \rightarrow \text{Rep } M(F_v)$ and $\text{Ind}_P^G: \text{Rep } P(F_v) \rightarrow \text{Rep } G(F_v)$ are adjoint.

Sketch. Note that

$$\text{Hom}_{G(F_v)}(V, \text{Ind}_P^G W) \cong \text{Hom}_{P(F_v)}(V, W) \cong \text{Hom}_{M(F_v)}(J(V), W).$$

The first isomorphism holds by “currying,” sending $f: V \rightarrow \text{Ind}_P^G W$ to $v \mapsto f(v)(1)$. The second isomorphism holds because the action of $N(F_v)$ on W is trivial. ■

Remark 50. The functor Ind_P^G is exact on smooth representations. We already have left-exactness by the adjunction, so the only issue is right-exactness. The main idea is to write

$$\text{Ind}_P^G(V) \cong \bigoplus_{s \in P(F_v) \backslash G(F_v) / K} V^{M(F_v) \cap s K s^{-1}},$$

where $K \subseteq G(F_v)$ is some compact open subgroup.

Lemma 51. Let V be a smooth representation of $N(F_v)$. Then

$$V(N) = \bigcup_{K \subseteq N(F_v)} \ker(R_K: V \rightarrow V),$$

where the union is over compact open subgroups $K \subseteq N(F_v)$, and R_K is the averaging operator.

Sketch. In one direction, note that $nv - v$ is in $\ker R_K$ for some K by choosing $K \subseteq N(F_v)$ large enough to contain n . Conversely, if $\int_K nv \, dn = 0$, then suppose that v is fixed by some compact open $K' \subseteq K$, and then $v \simeq \sum_{s \in K/K'} (sv - v)$. ■

Remark 52. The functor J is exact on smooth representations. We already have right-exactness. Left-exactness basically follows from the lemma, which explains why $-(N)$ is right-exact.

Lemma 53 (transitivity). Let $P \subseteq G$ be a parabolic with Levi decomposition $P = MN$, and let $P' \subseteq M$ be a parabolic with Levi decomposition $P' = M'N'$. Then

$$\mathrm{Ind}_P^G \circ \mathrm{Ind}_{P'}^{M'} = \mathrm{Ind}_{P'N}^G.$$

Sketch. By the adjunction, it is enough to check a corresponding statement on Jacquet modules, which holds basically because the unipotent radical of $P'N$ is $N'N$. ■

Proposition 54. Let $P \subseteq G$ be a parabolic with Levi decomposition $P = MN$. If V is an admissible representation of $G(F_v)$, then $J(V)$ is an admissible representation of $M(F_v)$.

Sketch. We use the Iwahori decomposition; set $M_n := M(F_v) \cap K_n$ and $N_n^\pm := N^\pm(F_v) \cap K_n$ for brevity. We will show that $J(V)^{M_n}$ is finite-dimensional for each n . By lifting finite-dimensional subspaces of $J(V)^{M_n}$ to V and conjugating, we see that $J(V)^{M_n}$ can be covered by (translates of) $J(V)^{M_n N_n^-}$.

Thus, it is enough to show that $V^{M_n N_n^-}$ and V^{K_n} have the same image in $J(V)$. To see the claim, note that the difficult inclusion will follow from the claim

$$R_{K_n}(v) \equiv v \pmod{\ker R_{N_n^+}}$$

for all $v \in V^{M_n N_n^-}$; one shows this by using the Iwahori decomposition to compute $R_{K_n}(v) = R_{N_n^+}(v)$. ■

2.3 Supercuspidal Representations

Definition 55 (supercuspidal, quasicuspidal). A representation V of $G(F_v)$ is *quasicuspidal* if and only if it is smooth, and its matrix coefficients have compact support modulo $Z(G(F_v))$. If it is further admissible, then V is *supercuspidal*.

Proposition 56. Let V be an irreducible quasicuspidal representation of $G(F_v)$. Then V is supercuspidal.

Sketch. We proceed by contraposition, supposing that V fails to be admissible, and then we will exhibit a matrix coefficient which is not compactly supported modulo $Z(G(F_v))$. We are given a compact open subgroup $K \subseteq G(F_v)$ for which V^K is infinite-dimensional; let $R_K: V \rightarrow V^K$ be the averaging operator.

Let's just describe the construction of the matrix coefficient. Fix some nonzero $\varphi \in V$, and then there is an infinite set for which $S \subseteq G(F_v)$ such that $\{R_K(g\varphi) : g \in S\}$ is a basis of V^K . Define $\varphi^\vee \in V^\vee$ by sending $\ker R_K$ to zero and sending $R_K(g\varphi)$ to 1 for each $g \in S$. Then $x \mapsto \langle x\varphi, \varphi^\vee \rangle$ is the requested matrix coefficient. ■

Definition 57 (algebraically cuspidal). A representation V of $G(F_v)$ is *algebraically cuspidal* if and only if $J_N(V) = 0$ for all N , as N varies over the unipotent radicals of parabolic subgroups of G .

Proposition 58. Let V be a smooth finitely generated representation of $G(F_v)$. Then there exists a parabolic subgroup $P \subseteq G$ with Levi decomposition $P = MN$ and some algebraically cuspidal smooth irreducible representation W of $M(F_v)$ for which there is a nonzero intertwining map $V \rightarrow \mathrm{Ind}_P^G W$.

Sketch. Let $P \subseteq G$ be a maximal (standard) parabolic admitting Levi decomposition $P = MN$ for which $J_N(V) \neq 0$; if no such proper parabolic exists, then V is already algebraically cuspidal, and there is nothing to do. Otherwise, by transitivity of the Jacquet module, $J_N(V)$ is algebraically cuspidal. Now, $J_N(V)$ is a finitely generated nontrivial representation of $M(F_v)$, so it admits a maximal subrepresentation by a Zorn's lemma argument, so it admits an irreducible quotient W . Note W is algebraically cuspidal as well, so we conclude by Frobenius reciprocity. ■

Remark 59. If V is actually irreducible, then the pair (M, W) is unique (up to suitable conjugation).

Example 60. It's worthwhile to explicate this for $G = \mathrm{GL}_2$. Let V be a smooth irreducible representation. If $J(V) = 0$, then V is already supercuspidal. Otherwise, $J(V)$ is nonzero and thus admits a one-dimensional quotient χ ; the map $J(V) \rightarrow \chi$ induces a nonzero intertwining map $V \rightarrow \mathrm{Ind}_B^G \chi$.

Theorem 61. Let V be a smooth representation of $G(F_v)$. Then V is algebraically cuspidal if and only if V is quasicuspidal.

Sketch. This essentially follows from the Cartan decomposition, so fix an Iwasawa decomposition $G(F_v) = P_0(F_v)K_0$ and maximal split torus $A_0 \subseteq P_0$. Let $A \subseteq Z(G)$ be a maximal split torus.

- In the forward direction, we would like to show that a chosen matrix coefficient $g \mapsto \langle g\varphi, \varphi^\vee \rangle$ is compactly supported modulo $Z(G)$. By the Cartan decomposition, it is enough to choose some $\alpha \in \Delta$ and show

$$\langle K_0 \varpi^\lambda K_0 \varphi, \varphi^\vee \rangle \stackrel{?}{=} 0$$

for $\lambda \in X_*(A_0)^+$ with $\alpha(\lambda)$ large.

Let's describe the construction. In short, set $P := P_{\Delta \setminus \{\alpha\}}$ with standard Levi decomposition $P = MN$, and then we may use that V is algebraically cuspidal to choose an open compact subgroup $N_\varphi \subseteq N(F_v)$ for which $R_{N_\varphi}(\varphi) = 0$. On the other hand, use smoothness to choose an open compact $N_{\varphi^\vee} \subseteq N(F_v)$ fixing every vector in $K_0 \cdot \varphi^\vee$. It is now enough to choose λ large enough so that $\varpi^\lambda N_\varphi \varpi^{-\lambda} \subseteq N_{\varphi^\vee}$.

- In the reverse direction, we would like to show that $V = V(N)$, where N is the unipotent radical of the standard parabolic $P := P_{\Delta \setminus \{\alpha\}}$. Well, choose $\varphi \in V$, and we want to show that $R_{N_\varphi}(\varphi) = 0$ for some compact open $N_\varphi \subseteq N(F_v)$.

The key idea is to test for $R_{N_\varphi}(\varphi) = 0$ against given linear functionals $\varphi^\vee \in V^\vee$, which turns out to be enough by a smoothness argument. Let's explain where the quasicuspidal hypothesis is used: by reversing the argument of the previous point, we see that

$$\langle \varpi^\lambda \varphi, K_0 \varphi^\vee \rangle = 0$$

for $\alpha(\lambda) \gtrsim_{\varphi^\vee} 1$. Then $N_{\varphi^\vee} := \varpi^{-\lambda}(K_0 \cap N(F_v))\varpi^\lambda$ has $\varphi^\vee(R_{N_{\varphi^\vee}}\varphi) = 0$. ■

Corollary 62. Let V be a smooth irreducible representation of $G(F_v)$. Then V is admissible.

Sketch. Embed V into the parabolic induction of an algebraically cuspidal irreducible representation. The latter is admissible. ■

Theorem 63 (Uniform admissibility). Fix a compact open subgroup K of $G(F_v)$. Then there is a constant $D(K)$ such that

$$\dim V^K \leq D(K)$$

for all smooth irreducible representations V of $G(F_v)$.

Sketch. Omitted for time. ■

2.4 Matrix Coefficients

Definition 64 (square integrable). Suppose G is reductive. A smooth representation V of $G(F_v)$ is *square integrable* if and only if its matrix coefficients are in $L^2(G(F_v))$; it is *essentially square integrable* if and only if its matrix coefficients are in $L^2(G(F_v)^1)$ upon restriction.

Definition 65 (tempered). Suppose G is reductive. A smooth representation V of $G(F_v)$ is *tempered* if and only if its matrix coefficients are in $L^{2+\varepsilon}(G(F_v))$ for all $\varepsilon > 0$; it is *essentially square integrable* if and only if its matrix coefficients are in $L^{2+\varepsilon}(G(F_v)^1)$ for all $\varepsilon > 0$ upon restriction.

Lemma 66 (Orthogonality). Suppose that V and W are admissible irreducible representations of $G(F_v)$ such that $V \not\cong W^\vee$ on $G(F_v)^1$, and suppose V is supercuspidal. For $v \in V$ and $v^\vee \in V^\vee$ and $w \in W$ and $w^\vee \in W^\vee$, we have

$$\int_{G(F_v)^1} \langle gv, v^\vee \rangle \langle gw, w^\vee \rangle dg = 0.$$

Sketch. The assumption that V is supercuspidal implies that the integrand is compactly supported. Fix $v^\vee$ and $w^\vee$. As a function in v and w , this map is a $G(F_v)$ -invariant pairing $V \times W \rightarrow \mathbb{C}$, which must vanish. ■

Lemma 67 (Orthogonality). Let V be a supercuspidal irreducible representation of $G(F_v)$. Then there is a constant $c_\pi > 0$ for which

$$\int_{G(F_v)^1} \langle gv, v^\vee \rangle \langle w, gw^\vee \rangle dg = c_\pi \langle v, w^\vee \rangle \langle w, v^\vee \rangle.$$

Sketch. Fixing $v^\vee$ and w shows that this must be proportional to $\langle v, w^\vee \rangle$. Similarly, that constant of proportionality must be proportional to $\langle v^\vee, w \rangle$. Positivity follows by plugging in some vectors. ■

Definition 68 (character). Let (π, V) be an admissible representation of $G(F_v)$. For any $f \in C_c^\infty(G(F_v))$, choose a compact open subgroup $K \subseteq G(F_v)$ for which f is bi-invariant under K ; then choose any finite-dimensional subspace W containing V^K and define

$$\mathrm{tr} \pi(f) := \mathrm{tr}(\pi(f); W).$$

This definition gives a well-defined *character* distribution $\mathrm{tr} \pi: C_c^\infty(G(F_v)) \rightarrow \mathbb{C}$.

Theorem 69 (Harish-Chandra). Suppose $\text{char } F = 0$. Let $G^{\text{rs}} \subseteq G$ be the Zariski open subset of regular semisimple elements. Then there is $\Theta_\pi \in C^\infty(G^{\text{rs}}(F_v))$ such that

$$\text{tr } \pi(f) = \int_{G(F_v)} \Theta_\pi(x) f(x) dx$$

for all $f \in C_c^\infty(G(F_v))$.

Sketch. Hard, so entirely omitted. ■

Proposition 70. Let $\{(\pi_i, V_i)\}_i$ be a finite collection of pairwise-non-isomorphic irreducible admissible representations of $G(F_v)$. Then the distributions $\{\text{tr } \pi_i\}_i$ are linearly independent.

Sketch. We pass to a finite-dimensional statement. Choose compact open $K \subseteq G(F_v)$ small enough so that $V_i^K \neq 0$ for each i . Now, let A be the image of the action map

$$\mathcal{H}_{G,K} \rightarrow \prod_i \text{End}_{\mathbb{C}}(V_i^K).$$

The finite-dimensional algebra A admits the $\{V_i\}_i$ as a finite set of irreducible modules for which A acts faithfully on the set. Thus, A is semisimple, and the proposition follows from orthogonality of characters for the irreducible modules of A . ■

Corollary 71. Suppose that G is split reductive, and let $\iota: G \rightarrow G$ be the Cartan involution. Let (π, V) be an irreducible admissible representation of $G(F_v)$. Then $V^\vee$ is isomorphic to the representation $g \mapsto \pi(\iota g)$.

Sketch. It is enough to compare the characters. Let $\tilde{V}$ be the latter representation. Fix $f \in C_c^\infty(G(F_v))$. On one hand, $\Theta_{V^\vee}(f) = \Theta_V(f \circ (-)^{-1})$ by rearranging the pairing. On the other hand, $\Theta_{\tilde{V}}(f) = \Theta_V(f \circ \iota)$. From here, one can check (on the Lie algebra) that g and $\iota(g)^{-1}$ are conjugate for any $g \in G(F_v)$. ■

Proposition 72. Let (π, V) be an irreducible supercuspidal representation of $G(F_v)$. Suppose $Z_G(F_v)$ is compact. For any $f \in C_c^\infty(G(F_v))$, there is $f_\pi \in C_c^\infty(G(F_v))$ such that

$$\pi'(f_\pi) = \begin{cases} \pi(f) & \text{if } \pi' \cong \pi, \\ 0 & \text{otherwise,} \end{cases}$$

for any irreducible admissible representation π' of $G(F_v)$.

Sketch. The argument present in Getz–Hahn has a gap. ■

Proposition 73. Let (π, V) be an irreducible supercuspidal representation of $G(F_v)$. Then there is $f_\pi \in C_c^\infty(G(F_v))$ with the following property: for any irreducible (V', π') , if $\text{tr } \pi'(f_\pi) \neq 0$, there is a character χ on $G(F_v)$ for which $\pi' \cong \pi \otimes \chi$.

Sketch. Let f_π be a matrix coefficient of V . Orthogonality implies that $\text{tr } \pi'(f_\pi) = 0$ unless V and V' are isomorphic as representations of $G(F_v)^1$. In this latter case, identify V and V' as vector spaces, and then one can show that $\chi(g) := \pi'(g)\pi(g)^{-1}$ factors through $G(F_v) \rightarrow G^{\text{ab}}(F_v)$, which is covered by the center, so χ is the desired character. ■

Remark 74. Characters of the principal series live in a continuous family, so one cannot single out the principal series with a finite collection of functions. One shows the claimed continuity by a direct calculation relating to the character of an induction to the original representation.

2.5 The Whittaker Model

Definition 75 (generic). Let G be a quasi-split reductive group over F_v . Let B be a Borel subgroup of G with unipotent radical N , and let $T \subseteq B$ be a maximal split torus. A character $\psi: N(F_v) \rightarrow \mathbb{C}^\times$ is *generic* if and only if $\psi|_{N_\alpha(F)} \neq 1$ for each $\alpha \in \Delta$.

Definition 76 (Whittaker functional). Let G be a quasi-split reductive group over F_v , and let B be a Borel subgroup with unipotent radical N . Choose $\psi: N(F_v) \rightarrow \mathbb{C}^\times$.

- If v is finite, a ψ -Whittaker functional is an element of $\text{Hom}_{N(F_v)}(V, \psi)$.
- If v is infinite, a ψ -Whittaker functional is a continuous linear functional in $\text{Hom}_{N(F_v)}(V_{\text{sm}}, \psi)$, where V_{sm} is given its natural Fréchet topology.

Remark 77. In the finite case, a ψ -Whittaker functional exists if and only if $J_\psi(V)$ admits a functional.

Definition 78 (generic). Let G be a quasi-split reductive group over F_v . A smooth irreducible representation V is *generic* if and only if it admits a ψ -Whittaker functional for some generic ψ .

Theorem 79. Let G be a quasi-split reductive group over F_v , and let B be a Borel subgroup with unipotent radical N . If $\psi: N(F_v) \rightarrow \mathbb{C}^\times$ is generic, the space of ψ -Whittaker functionals of any smooth irreducible representation of V is at most one.

Sketch. We only sketch this when G is split and v is finite. Let $\mathcal{H}' \subseteq \mathcal{H}_G$ be the eigenvectors of $N(F) \times N(F)$ with eigenvalue $\psi_N \boxtimes \psi_N^{-1}$. By a computation with the Bruhat decomposition, one finds that the Cartan involution ι acts trivially on $\mathcal{H}'$; this uses the hypothesis that ψ is generic.

Now, choose two nonzero ψ -Whittaker functionals $\lambda_1, \lambda_2: V \rightarrow \mathbb{C}$. It is enough to show that $\ker \lambda_1 = \ker \lambda_2$. By identifying V with $V^\vee$ (as vector spaces with different $G(F_v)$ -actions), we see that λ_i becomes a ψ^{-1} -Whittaker functional on $V^\vee$. Now, the distributions

$$\varphi \mapsto \langle \lambda_1 * \varphi, \lambda'_1 \rangle \quad \text{and} \quad \varphi \mapsto \langle \lambda_2 * \varphi, \lambda'_2 \rangle$$

both live in $\mathcal{H}'$, and a direct calculation shows that they differ by the action of ι , so they must actually be equal! This can be rearranged into the desired result. ■

Definition 80 (Whittaker function). Let G be a quasi-split reductive group over F_v , and let B be a Borel subgroup with unipotent radical N . For any generic $\psi: N(F_v) \rightarrow \mathbb{C}^\times$, a ψ -Whittaker function is an element of the space $\mathcal{W}(\psi) := C^\infty((N(F_v), \psi) \backslash G(F_v))$.

Remark 81. By Frobenius reciprocity, the presence of a ψ -Whittaker functional λ of V is equivalent to admitting a Whittaker model $V_{\text{sm}} \rightarrow \mathcal{W}(\psi)$ (given by sending v to $W_v: g \mapsto \lambda(gv)$).

Theorem 82. Every supercuspidal representation of $\mathrm{GL}_n(F_v)$ is generic.

Sketch. We will show this for $n = 2$ shortly. ■

Proposition 83. Any non-generic smooth representation V of $\mathrm{GL}_2(F_v)$ factors through $\det$.

Sketch. This is pretty tricky. Consider the action $\hat{\pi}$ of $C_c^\infty(F)$ on V by

$$\hat{\pi}(f)v := \int_F \hat{f}(x) \cdot \begin{bmatrix} 1 & -x \\ & 1 \end{bmatrix} v \, dx,$$

where the Fourier transform is taken with respect to some fixed $\psi: F_v \rightarrow \mathbb{C}^\times$. This action is (co)smooth, and $1_{a+\mathfrak{p}^k}$ acts by $\psi_a 1_{\mathfrak{p}^k}$ (up to scalar). Now, define a sheaf $\mathcal{F}_V$ on F on the base of compact open subsets given by $U \mapsto 1_U V$. A direct calculation shows that the stalk of $\mathcal{F}_V$ at $a \in F$ is $J_{\psi_a}(V)$, so $\mathcal{F}_V$ must be a skyscraper sheaf! Thus, $V = J(V)$, so $N(F_v)$ acts trivially on V . Taking conjugates shows that $\mathrm{SL}_2(F_v)$ acts trivially on V . ■

3 Representation Theory of GL_2

3.1 The Principal Series

Lemma 84. Let P be a parabolic subgroup of G with Levi decomposition $P = MN$. For any character $\chi: M(F_v) \rightarrow \mathbb{C}^\times$, the map $R: C_c^\infty(G(F_v)) \rightarrow \mathrm{Ind}_P^G \chi$ defined by

$$R_\varphi(x) := \int_{P(F_v)} \varphi(p^{-1}x) \delta^{1/2} \chi(x) \, dx$$

is surjective.

Sketch. Fix an Iwasawa decomposition $G(F_v) = P(F_v)K$. Given $f \in \mathrm{Ind}_P^G \chi$, define $\varphi: G(F_v) \rightarrow \mathbb{C}$ by $\varphi := f 1_K$. Then $R_\varphi \simeq f$. ■

Lemma 85. Fix a character $\xi: G(F_v) \rightarrow \mathbb{C}^\times$. Any eigen-distribution $D: C_c^\infty(G(F_v)) \rightarrow \mathbb{C}$ satisfying $D(L_g^{-1}f) = \bar{\xi}(g)D(f)$ is proportional to the distribution

$$f \mapsto \int_{G(F_v)} \xi(h) f(h) \, d_\ell h.$$

Sketch. By twisting D , we may assume $\xi = 1$. Now, fix a compact open subgroup $K \subseteq G(F_v)$, and normalize $d_\ell h$ so that K has volume one. Then the uniqueness of the left-invariant Haar measure implies that

$$Tf = T(1_K) \int_{G(F_v)} f(h) \, d_\ell h,$$

as required. ■

Proposition 86. Let G be a split reductive group with Borel $B = TN$. For each character $\chi: T(F_v) \rightarrow \mathbb{C}^\times$ and generic $\psi: N(F_v) \rightarrow \mathbb{C}^\times$,

$$\dim J_\psi \left(\text{Ind}_B^G \chi \right) = 1.$$

Sketch. This is basically Mackey theory. To start, note that

$$f \mapsto \int_{N(F_v)} f(wn) \psi(n)^{-1} dn$$

provides the required functional. (Even if f does not have compact support, one can regularize the integral because f will be constant “at ∞ ,” so the integral against ψ^{-1} vanishes.) It remains to upper-bound. Via the projection $C_c^\infty(G(F_v)) \rightarrow \text{Ind}_B^G \chi$, it is enough to show that

$$\dim \left\{ D \in C_c^\infty(G(F_v))^\vee : L_b D = \delta^{1/2} \chi(b) D \text{ and } R_n D = \psi(n)^{-1} D \right\} \leq 1.$$

To avoid a complicated induction on the Bruhat decomposition, we only show this for $G = \text{GL}_2$. Then the Bruhat decomposition simply gives

$$0 \rightarrow C_c^\infty(B(F_v))^\vee \rightarrow C_c^\infty(\text{GL}_2(F_v))^\vee \rightarrow C_c^\infty(B(F_v)wN(F_v))^\vee \rightarrow 0,$$

where w is the long Weyl element. Now, any suitable eigen-distribution D is uniquely determined up to scalar when restricted to $B(F_v)wN(F_v) \cong B(F_v) \times N(F_v)$. Thus, any two suitable eigen-distributions are equal up to an eigen-distribution from $B(F_v)$; but there are no suitable eigendistributions on $B(F_v)$, which we see by checking that the previous lemma does not provide a suitable eigen-distribution if we just use one of the eigen-conditions. ■

Remark 87. The same argument shows that $\dim J(\text{Ind}_B^G \chi) \leq \text{rank } G$, and I think equality holds. For $G = \text{GL}_2$, one can show by hand that the functionals $f \mapsto f(1)$ and

$$f \mapsto \int_{F_v} \left(f \left(w \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \right) - |x|^{-1} \chi_1^{-1} \chi_2(x) 1_{x \notin \mathcal{O}_v} \varphi(1) \right) dx$$

are both well-defined and linearly independent.

Proposition 88. Let G be a split reductive group with Borel $B = TN$. If $\chi: T(F_v) \rightarrow \mathbb{C}^\times$ has trivial stabilizer via the $(W, \cdot)$ -action, then $\text{Ind}_B^G \chi$ is irreducible.

Sketch. We only show this for $G = \text{GL}_2$. Suppose that $V := \text{Ind}_B^G \chi$ lives in a short exact sequence

$$0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0.$$

For dimension reasons, $J_\psi(V') = 0$ or $J_\psi(V'') = 0$. By passing to the contragredient, we may take $J_\psi(V') = 0$. Now, V' factors through $\det$, so V' admits an eigenvector, so V admits a one-dimensional subspace; let f be an eigenvector. Then f satisfies

$$f(bg) = \delta^{1/2} \chi(b) \rho(\det g) f(1)$$

for some character $\rho: F^\times \rightarrow \mathbb{C}^\times$. Plugging in $b = g^{-1} = \text{diag}(a, 1/a)$ reveals that $\chi_1 \chi_2^{-1} = |\cdot|^{-1}$, so χ is fixed by the $(W, \cdot)$ -action. ■

Remark 89. Even in the exceptional cases $\chi_1\chi_2^{-1} = |\cdot|^{-1}$, the same argument shows that $\text{Ind}_B^G \chi$ contains an eigenvector with irreducible quotient. Dually, when $\chi_1\chi_2^{-1} = |\cdot|$, we see $\text{Ind}_B^G \chi$ contains a one-dimensional quotient with irreducible kernel.

Proposition 90. Let G be a split reductive group with Borel $B = TN$. If $\chi, \chi': T(F_v) \rightarrow \mathbb{C}^\times$ have $\text{Ind}_B^G \chi \cong \text{Ind}_B^G \chi'$, then $\chi' \in W\chi$.

Sketch. This is Mackey theory. Again, to avoid complicated inductions, we only show this for $G = \text{GL}_2$. Fix a nonzero morphism $\text{Ind}_B^G \chi \rightarrow \text{Ind}_B^G \mu$. By Frobenius reciprocity, we get a nonzero $B(F_v)$ -invariant map $\text{Ind}_B^G \chi \rightarrow \delta^{1/2} \mu$. Via the projection $C_c^\infty(G(F_v)) \rightarrow \text{Ind}_B^G \chi$, we receive a distribution D satisfying

$$\begin{cases} L_b D = \delta^{-1/2} \chi(b) D, \\ R_b D = \delta^{-1/2} \mu(b)^{-1} D, \end{cases}$$

for $b \in B(F_v)$. If D is nonzero on $B(F_v)wN(F_v) \cong B(F_v) \times N(F_v)$, then the eigenvector conditions on $B(F_v) \times N(F_v)$ give D an explicit formula, and then the remaining eigenvector condition on $T(F_v)$ forces $\mu = {}^w\chi$. Otherwise D is supported on $B(F_v)$, and one can compare the two eigenvector conditions to get $\mu = \chi$. ■

Proposition 91. Let G be a split reductive group with Borel $B = TN$, and choose $\chi: T(F_v) \rightarrow \mathbb{C}^\times$. If χ has trivial stabilizer via the W -action, then

$$J(\text{Ind}_B^G \chi) \cong \delta^{1/2} \otimes \bigoplus_{w \in W} {}^w\chi.$$

Sketch. For a character $\mu: T(F_v) \rightarrow \mathbb{C}^\times$, Frobenius reciprocity gives

$$\text{Hom}_{T(F_v)}(J(\text{Ind}_B^G \chi), \delta^{1/2} \mu) \cong \text{Hom}_{G(F_v)}(\text{Ind}_B^G \chi, \text{Ind}_B^G \mu).$$

By the previous proposition, this is nonzero exactly for $\mu \in W\chi$, so we have a quotient isomorphic to $\delta^{1/2} \otimes \bigoplus_{w \in W} {}^w\chi$. This is an isomorphism for dimension reasons. ■

Example 92. For $G = \text{GL}_2$, let's describe what happens in the case $\chi = {}^w\chi$. Then $J(\text{Ind}_B^G \chi)$ admits a unique quotient map to $\delta^{1/2} \chi$ (up to scalar); let the kernel be $\delta^{1/2} \mu$, so there is a short exact sequence

$$0 \rightarrow \delta^{1/2} \mu \rightarrow J(\text{Ind}_B^G \chi) \rightarrow \delta^{1/2} \chi \rightarrow 0.$$

If $\chi \neq \mu$, this splits, which is a contradiction. Instead, we must have $\mu = \chi$, and the short exact sequence must be non-split. Some explicit calculations on bases shows that we can give $J(\text{Ind}_B^G \chi)$ a basis so that it is isomorphic to

$$t \mapsto \delta^{1/2} \chi(t) \begin{bmatrix} 1 & v(t_1/t_2) \\ & 1 \end{bmatrix}.$$

3.2 Unramified Representations

Theorem 93. Let G be a split reductive group, and let $K \subseteq G(F_v)$ be a maximal compact subgroup. Then $\mathcal{H}_K$ is commutative.

Sketch. Use Gelfand's trick. The relevant anti-involution is given by the (inverse of the) Cartan involution. For example, if $G = \mathrm{GL}_n$, then the anti-involution is $(-)^{-\mathrm{T}}$. ■

Definition 94. Fix a split reductive group G with maximal compact subgroup $K \subseteq G(F_v)$. A smooth representation V is *spherical* if and only if $V^K \neq 0$.

Remark 95. If V is spherical and irreducible, then V^K is an irreducible $\mathcal{H}_K$ -module, so $\dim V^K = 1$.

Theorem 96 (Satake). Let G be a split reductive group with Borel $B = TN$ and Iwasawa decomposition $G(F_v) = B(F_v)K$. Let $\widehat{T}$ be the Langlands dual of T . Then

$$\mathcal{H}_K \cong \mathcal{O}(\widehat{T})^W.$$

Sketch. We only argue this for $G = \mathrm{GL}_2$ with $K = \mathrm{GL}_2(\mathcal{O}_v)$, and we will argue some of this now. For each $k \geq 0$, let $T_k \in \mathcal{H}_K$ be the indicator of $g \in M_2(\mathcal{O})$ with $(\det g)\mathcal{O} = \mathfrak{p}^k$; set $T := T_1$ for brevity. Also, define R to indicate ϖK . An induction shows that $\mathcal{H}_K$ is spanned by the vectors $R^a T_k$ where $a \in \mathbb{Z}$ and $k \geq 0$. By examining actions on $\mathbb{P}_{\mathbb{F}_q}^1$, one can show that

$$K \operatorname{diag}(\varpi, 1) K = \begin{bmatrix} \varpi & \\ & 1 \end{bmatrix} K \sqcup \bigsqcup_{b \in \mathbb{F}_q} \begin{bmatrix} \varpi & b \\ & 1 \end{bmatrix} K.$$

From here, some explicit calculation shows that $T_1 T_k = T_{k+1} + q R T_{k-1}$ for $k \geq 1$. It follows that the natural map $\mathbb{C}[T, R, 1/R] \rightarrow \mathcal{H}_K$ is a surjection, and this surjection will soon be checked to be an isomorphism. The source is isomorphic to $\mathbb{C}[t_1 + t_2, t_1 t_2, 1/t_1 t_2]$, which we call good enough. ■

Example 97. Let $\alpha: F_v^\times \rightarrow \mathbb{C}^\times$ be an unramified character. Then α is spherical. For $G = \mathrm{GL}_2$, a direct calculation shows that T_1 acts by $(q+1)\alpha(\varpi)$ and R acts by $\alpha(\varpi)^2$.

Example 98. Let $\chi: T(F_v) \rightarrow \mathbb{C}^\times$ be an unramified character. By the Iwasawa decomposition, $\operatorname{Ind}_B^G \chi$ is unramified with the unique unramified vector $f: G(F_v) \rightarrow \mathbb{C}$ given by $f(bk) = \delta^{1/2} \chi(b)$. For $G = \mathrm{GL}_2$, a direct calculation shows that T_1 acts by $q^{1/2}(\chi_1(\varpi) + \chi_2(\varpi))$ and R acts by $\chi_1(\varpi)\chi_2(\varpi)$.

Theorem 99. Let G be a split reductive group with Borel $B = TN$. Then every irreducible spherical representation V is a subquotient of $\operatorname{Ind}_B^G \chi$, where $\chi: T(F_v) \rightarrow \mathbb{C}^\times$ is an unramified character (unique up to Weyl orbit from V).

Sketch. The fact that $\operatorname{Ind}_B^G \chi$ admits a unique irreducible spherical subquotient follows from the fact that $\dim(\operatorname{Ind}_B^G \chi)^K = 1$. Now, the irreducible spherical representations embed into the characters of $\mathcal{H}_K$, so it merely remains to check that subquotients of $\operatorname{Ind}_B^G \chi$ exhausts all such characters; this is an application of the Satake isomorphism.

Let's explicate this for $G = \mathrm{GL}_2$. Any character $\xi: \mathcal{H}_K \rightarrow \mathbb{C}$ is uniquely determined by $\xi(T)$ and $\xi(R)$, and we must have $\xi(R) \in \mathbb{C}^\times$. Let α_1 and α_2 be the roots of the polynomial

$$X^2 - q^{-1/2} \xi(T) X + \xi(R) = 0.$$

Then define $\chi_i: F^\times \rightarrow \mathbb{C}^\times$ to be the unique unramified character for which $\chi_i(\varpi) = \alpha_i$. Then the character of $\mathcal{H}_K$ induced by $\text{Ind}_B^G \chi$ agrees with the character ξ . The only problem occurs when $\text{Ind}_B^G \chi$ fails to be irreducible; up to symmetry, this means $\chi_1 \chi_2^{-1} = |\cdot|^{-1}$. Then ξ is induced by the unramified character $\alpha: F_v^\times \rightarrow \mathbb{C}^\times$ defined by $\alpha(\varpi) = q^{1/2} \alpha_2$. ■

Remark 100. This argument completes the proof of Satake's isomorphism for $G = \text{GL}_2$: indeed, it shows that the ring surjection $\mathbb{C}[T, R, 1/R] \rightarrow \mathcal{H}_K$ induces a surjective closed embedding $\text{Spec } \mathcal{H}_K(\mathbb{C}) \rightarrow \text{Spec } \mathbb{C}[T, R, 1/R](\mathbb{C})$ with reduced target.

Theorem 101 (Casselman–Shalika). Let G be a split reductive group with Borel $B = TN$. Choose $\lambda \in X^*(T)_\mathbb{C}$, and define $J(\lambda)$ to be the unique spherical irreducible quotient of

$$\text{Ind}_B^G e^{\langle H_T(-), \lambda \rangle}.$$

Choose a generic character ψ , and suppose W is a spherical vector in the Whittaker model of $J(\lambda)$.

- (a) If $W \neq 0$, then $W(1) \neq 0$.
- (b) If $\mu \in X^*(T) \setminus X^*(T)_+$, then $W(\varpi^\mu) = 0$.
- (c) If $\mu \in X^*(T)_+$, then

$$W(\varpi^\mu) = \delta_B^{1/2}(\varpi^\mu) \chi_\mu(q^{-\lambda}) W(1).$$

Sketch. We only argue this for $G = \text{GL}_2$, for which we give an ad-hoc argument. Let the representation be $\text{Ind}_B^G \chi$. We are tasked with computing the numbers $w_n := W(\text{diag}(\varpi^n, 1))$; the Iwasawa decomposition shows that W is uniquely determined by these values. Set $\beta_i := q^{-1/2} \chi_i(\varpi)$ for brevity. Using $W\left(\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g\right) = \psi(x)W(g)$, one can show that $w_n = 0$ if $n < 0$. We know that $T \cdot W = q(\beta_1 + \beta_2)W$. Comparing this with a direct expansion of $T \cdot W$ gives the recursion

$$w_{n+2} = (\beta_1 + \beta_2)w_{n+1} - \beta_1 \beta_2 w_n.$$

Solving this recursion gives $w_n = (\beta_1^{n+1} - \beta_2^{n+1}) / (\beta_1 - \beta_2) \cdot w_0$ and hence the result. ■

3.3 The Kirillov Model

Definition 102 (Kirillov model). Fix a generic smooth representation V of $\text{GL}_2(F_v)$, and let $W_\bullet: V \rightarrow C^\infty((N(F_v), \psi) \backslash \text{GL}_2(F_v))$ be a Whittaker model. Then the *Kirillov model* consists of the subspace of $C_c^\infty(F^\times)$ of functions

$$\varphi_v(x) := W_v(\text{diag}(x, 1)).$$

Remark 103. For φ in the Kirillov model, one has

$$\begin{bmatrix} 1 & b \\ & 1 \end{bmatrix} \varphi(x) = \psi(bx) \varphi(x) \quad \text{and} \quad \begin{bmatrix} a & \\ & d \end{bmatrix} \varphi(x) = \omega(d) \varphi(ax/d),$$

where ω is the central character.

Lemma 104. Fix a generic smooth representation V of $\text{GL}_2(F_v)$, which we identify with its Kirillov model. For any $\varphi \in V$, we have $\varphi(y) = 0$ if $|y| \gtrsim_\varphi 1$.

Sketch. By smoothness, φ is stabilized by some subgroup $\{\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} : x \in \mathfrak{p}^n\}$. Thus, for $x \in \mathfrak{p}^n$, we see $(1 - \psi(xy))\varphi(y) = 0$, so $\varphi(y) = 0$ as soon as $y\mathfrak{p}^n \not\subseteq \ker \psi$. ■

Lemma 105. Fix a generic irreducible smooth representation V of $\mathrm{GL}_2(F_v)$, which we identify with its Kirillov model. The kernel of $V \rightarrow J(V)$ is exactly $C_c^\infty(F_v^\times)$.

Sketch. For one inclusion, recall $V(N)$ is spanned by vectors of the form $\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \varphi - \varphi$, so it is enough to show that such functions vanish on small $y \in F_v^\times$. This follows because $\psi(xy) = 1$ for small y .

For the other inclusion, because the kernel is nonzero, it is enough to show that $C_c^\infty(F_v^\times)$ is an irreducible as a $B(F_v)$ -module. Let $V \subseteq C_c^\infty(F_v^\times)$ be a nonzero subspace, and we will check that V contains sufficiently small indicators of any given $a \in F_v^\times$. By translation, we may assume there is $\varphi \in V$ with $\varphi(a) \neq 0$. Then a direct expansion shows that some $f \in C_c^\infty(N(F_v))$ has

$$f * \varphi = \widehat{f} \varphi,$$

so just choose $\widehat{f}$ to indicate a small open neighborhood of a . ■

Proposition 106. Fix a generic irreducible smooth representation V of $\mathrm{GL}_2(F_v)$, which we identify with its Kirillov model.

- (a) If $V = \mathrm{Ind}_B^G \chi$ is an irreducible principal series with $\chi_1 \neq \chi_2$, then V contains $|\cdot|^{1/2} \chi_1 1_{\mathcal{O}}$ and $|\cdot|^{1/2} \chi_2 1_{\mathcal{O}}$.
- (b) If $V = \mathrm{Ind}_B^G \chi$ is an irreducible principal series with $\chi_1 = \chi_2$, then V contains $|\cdot|^{1/2} \chi_1 1_{\mathcal{O}}$ and $v(t) |\cdot|^{1/2} \chi_1 1_{\mathcal{O}}$.
- (c) If $V \subsetneq \mathrm{Ind}_B^G \chi$ is the codimension-one subspace when $\chi_1 \chi_2^{-1} = |\cdot|$, then V contains $|\cdot|^{1/2} \chi_1 1_{\mathcal{O}}$.

Along with $C_c^\infty(F_v^\times)$, these vectors span V .

Sketch. We will only sketch (a) because the others are similar. Note $V(N) = C_c^\infty(F_v^\times)$, so it only remains to lift $J(V)$, which we recall is the representation $\delta^{1/2} \chi_1 \oplus \delta^{1/2} \chi_2$ of $T(F_v)$.

Choose $\varphi \in V$ such that $t\bar{\varphi} = \delta^{1/2} \chi(t) \bar{\varphi}$ in $J(V)$. We will show that $\varphi \simeq |\cdot|^{1/2} \chi_1$ for small t , which completes the argument. This is an intricate compactness argument. Well, given $t \in F^\times$, the lemma implies that

$$\varphi(tu) = \delta^{1/2} \chi(\mathrm{diag}(t, 1)) \varphi(u)$$

holds for small u ; say $|u| \leq \varepsilon(t)$. By smoothness, $\varepsilon(t)$ can taken to be locally constant, so we may define $\varepsilon := \min\{\varepsilon(t) : t \in \varpi \mathcal{O}^\times\}$. ■

3.4 Local Rankin–Selberg for GL_2

Definition 107 (local zeta integral). For $\varphi \in C^\infty(F^\times)$, define

$$Z(s, \varphi) := \int_{F^\times} \varphi(y) |y|^{s-1/2} d^\times y,$$

provided that the integral converges. For a character $\xi: F^\times \rightarrow \mathbb{C}^\times$, define $Z(s, \varphi, \xi) := Z(s, \varphi \xi)$.

Remark 108. We claim that $\{Z(s, \varphi) : \varphi \in C_c^\infty(F_v^\times)\} = \mathbb{C}[q^s, q^{-s}]$. The main point is that $Z(s, 1_{1+\mathfrak{p}^n})$ is a (positive) constant if $n > 0$, and $Z(s, \mathrm{diag}(a, 1)\varphi) = |a|^{1/2} |a|^{-s} Z(s, \varphi)$ for any $a \in F^\times$.

Definition 109 (local L -factor). Let V be a smooth generic irreducible representation of $\mathrm{GL}_2(F_v)$.

- If V is supercuspidal, define $L(s, \pi) := 1$.
- If $V = \mathrm{Ind}_B^G \chi$ is an irreducible principal series, define α_i to be $\chi_i(\varpi)$ if χ_i is unramified and zero otherwise; then define $L(s, \pi) := (1 - \alpha_1 q^{-s})^{-1} (1 - \alpha_2 q^{-s})^{-1}$.
- If $V \subsetneq \mathrm{Ind}_B^G \chi$ is the codimension-one subspace when $\chi_1 \chi_2^{-1} = |\cdot|$; then define $L(s, \pi) := (1 - \alpha_1 q^{-s})^{-1}$ with α_1 as above.

For a character $\xi: F^\times \rightarrow \mathbb{C}^\times$, define $L(s, \pi, \xi) := L(s, \pi \otimes \xi)$.

Proposition 110. Let V be a smooth generic irreducible representation of $\mathrm{GL}_2(F_v)$, which we identify with its Kirillov model. Then

$$\{Z(s, \varphi) : \varphi \in V\} = L(s, \pi) \mathbb{C}[q^s, q^{-s}].$$

In particular, $Z(s, \varphi)$ admits meromorphic continuation.

Sketch. There is nothing to say in the supercuspidal case. We only sketch the case of $V = \mathrm{Ind}_B^G \chi$ because the others are similar. Because $C_c^\infty(F_v^\times) \subseteq V$ yields $\mathbb{C}[q^s, q^{-s}]$, it remains to compute $Z(s, |\cdot|^{1/2} \chi_i 1_{\mathcal{O}})$. This is

$$\int_{\mathcal{O}} \chi_i(y) |y|^s d^\times y = \left(\sum_{k=0}^{\infty} \chi_i(\varpi)^k q^{-sk} \right) \int_{\mathcal{O}^\times} \chi_i(t) d^\times t.$$

The integral vanishes if χ_i fails to be unramified; otherwise, it is a positive constant. The infinite sum is $(1 - \chi_i(\varpi) q^{-s})^{-1}$. ■

Lemma 111 (Multiplicity one). Let V be a smooth generic irreducible representation of $\mathrm{GL}_2(F_v)$. Embed $F^\times \rightarrow T(F)$ by the first coordinate. Given $\chi: F^\times \rightarrow \mathbb{C}^\times$, there are at most two $s \in \mathbb{C}/2\pi i \log q \mathbb{Z}$ for which

$$\dim_{F^\times}(V, \chi |\cdot|^s) > 1.$$

Sketch. Suppose $s \in \mathbb{C}/2\pi i \log q \mathbb{Z}$ features linearly independent functionals $\ell_1, \ell_2: V \rightarrow \chi |\cdot|^s$. When restricted to $C_c^\infty(F^\times)$ in the Kirillov model, ℓ_1 and ℓ_2 are eigen-distributions and thus unique up to scalar, so there is nonzero $\ell \in \mathrm{span}\{\ell_1, \ell_2\}$ which vanishes on $V(N)$. But then ℓ realizes $\chi |\cdot|^s$ as a one-dimensional quotient of $J(V)$, which occurs for at most two s . ■

Theorem 112 (Functional equation). Let V be a smooth generic irreducible representation of $\mathrm{GL}_2(F_v)$, which we identify with its Kirillov model. Let ω be the central character. Then there is a meromorphic function $\gamma(s, \pi, \psi)$ such that

$$Z(1-s, w\varphi, \omega^{-1}) = \gamma(s, \pi, \psi) Z(s, \varphi).$$

Sketch. Being meromorphic follows from plugging in a particular φ to make $Z(s, \varphi)$ nonzero. Thus, the main issue is independence of φ : we will check that the two linear functionals

$$\varphi \mapsto Z(s, \varphi) \quad \text{and} \quad \varphi \mapsto Z(1-s, w\varphi, \omega^{-1})$$

are proportional for cofinitely many s . Both functionals provide $F^\times$ -equivariant maps to $|\cdot|^{-s+1/2}$, so this follows from the multiplicity one result. ■

3.5 Local Intertwining Operators

Definition 113. Let G be a split reductive group with Borel $B = TN$. For $w \in W$ and character $\chi: T(F_v) \rightarrow \mathbb{C}^\times$, define the intertwining operator $Mw: \text{Ind}_B^G \chi \rightarrow \text{Ind}_B^G {}^w\chi$ by

$$M_\chi w(\varphi)(g) := \int_{N(F_v)} \varphi(wng) dn,$$

provided the integral converges. We will usually abbreviate $M_\chi w$ to Mw or M_χ or M .

Remark 114. Direct calculation shows that M_χ is adjoint to $M_{w^{-1}\bar{\chi}^{-1}}$. Indeed, for suitable φ_1 and φ_2 , one has

$$\int_K Mw(\varphi_1)(x) \overline{\varphi_2(x)} dx = \int_K \int_{N(F_v)} \varphi_1(wnx) \overline{\varphi_2(nx)} dn dx = \int_{Z(F_v) \backslash G(F_v)} \varphi(wg) \overline{\psi(g)} dg.$$

Now apply $g \mapsto w^{-1}g$ and work backwards.

Proposition 115. Let G be a split reductive group with Borel $B = TN$. For $w \in W$, the intertwining operator $Mw: \text{Ind}_B^G \chi \rightarrow \text{Ind}_B^G {}^w\chi$ converges absolutely and uniformly in a suitable cone of $\chi \in X^*(T)_\mathbb{C}$.

Sketch. We only argue this for $G = \text{GL}_2$ and v finite. Then we are interested in the convergence of

$$Mw(\varphi)(g) = \int_{F_v} \varphi \left(w \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g \right) dx.$$

The region $|x| < 1$ is compact and gives no issue. The region $|x| \geq 1$ is bounded using the “by-hand” Iwasawa decomposition

$$\begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} = \begin{bmatrix} 1/x & -1 \\ & x \end{bmatrix} \begin{bmatrix} 1 & \\ 1/x & 1 \end{bmatrix}.$$

One then factors out $\int_{|x| \leq 1} |x|^{2\sigma-2} dx$, where $\sigma \in \mathbb{R}$ is chosen so that $|\chi_1 \chi_2^{-1}| = |\cdot|^\sigma$. Thus, we receive convergence when $\sigma > 1/2$. ■

Definition 116. Define the vector bundle $\int_{X^*(T)_\mathbb{C}}^\oplus \text{Ind}_B^G \chi d\chi$ over $X^*(T)_\mathbb{C}$ to be the locally trivial fiber bundle whose fiber over χ is $\text{Ind}_B^G \chi$. Here the trivialization $\text{Ind}_B^G \chi \rightarrow \text{Ind}_B^G \chi |\cdot|^s$ sends φ to

$$\varphi_s(g) := m_B(g)^{s\delta} \varphi(g),$$

where $m_B(g)$ is the Levi part of the Iwasawa decomposition of g .

Definition 117. Take $G = \text{GL}_2$. Fix a character $\chi: T(F_v) \rightarrow \mathbb{C}^\times$. For a Schwartz function $\Phi: F_v \times F_v \rightarrow \mathbb{C}$, define

$$\varphi_\Phi(g; \chi) := \frac{\chi_1 |\cdot|^{1/2} (\det g)}{L(1, \chi_1 \chi_2^{-1})} \int_{F_v^\times} \Phi(te_2 g) \chi_1 \chi_2^{-1}(t) |t| d^\times t.$$

The integral is defined via analytic continuation by realizing that it is a local Z -function in the sense of Tate’s thesis. Note φ_Φ defines a smooth global section of $\int_{X^*(T)_\mathbb{C}}^\oplus \text{Ind}_B^G \chi d\chi$.

Lemma 118. Take $G = \mathrm{GL}_2$. Every $\varphi \in \mathrm{Ind}_B^G \chi$ admits some Schwartz $\Phi_\varphi: F_v \times F_v \rightarrow \mathbb{C}$ for which $\varphi = \varphi_{\Phi_\varphi}$.

Sketch. Define

$$\bar{\Phi}_\varphi(v) := \begin{cases} \chi_1^{-1}(\det g) f(g) & \text{if } v = e_2 g, \\ 0 & \text{otherwise.} \end{cases}$$

This is well-defined. A direct calculation shows that $\varphi \simeq \varphi_{\bar{\Phi}_\varphi}$, where the constant is given by the measure of $\mathcal{O}^\times$ and $L(1, \chi_1 \chi_2^{-1})$. Rescaling provides Φ_φ . ■

Proposition 119. Take $G = \mathrm{GL}_2$, and fix a character $\chi: T(F_v) \rightarrow \mathbb{C}^\times$ in a suitable cone. For Schwartz $\Phi: F_v \times F_v \rightarrow \mathbb{C}$, one has

$$Mw(\varphi_\Phi) = \left(\frac{L(0, \chi_1 \chi_2^{-1})}{\varepsilon(\chi_1 \chi_2^{-1}; \psi) L(1, \chi_1 \chi_2^{-1})} \right) \cdot \chi_1 \chi_2(-1) \cdot \frac{\chi_2 |\cdot|^{1/2}(\det g)}{L(1, \chi_2 \chi_1^{-1})} \int_{F_v^\times} \mathcal{F}_\psi \Phi(te_2 g) \chi_1^{-1} \chi_2(t) |t| d^\times t,$$

where $\mathcal{F}_\psi \Phi$ is the Fourier transform with respect to $\psi: F_v \rightarrow \mathbb{C}^\times$.

Sketch. This is a long calculation. We remark that

$$\mathcal{F}_\psi \Phi(w) = \int_{F_v^2} \Phi(v) \psi(\det [v \ w]) dv.$$

Now, the main point is to use Tate's thesis to "flip" the character in the integral from $\chi_1 \chi_2^{-1}$ to $\chi_2^{-1} \chi_1$. In particular, one must compute the Fourier transform of $t \mapsto \Phi(te_2 g)$, which is $t \mapsto \mathcal{F}_\psi \Phi(te_2 g)$. ■

Theorem 120. Let G be a split reductive group with Borel $B = TN$. For $w \in W$, the intertwining operator Mw admits meromorphic continuation to a meromorphic operator

$$\int_{X^*(T)_\mathbb{C}}^{\oplus} \mathrm{Ind}_B^G \chi d\chi \rightarrow \int_{X^*(T)_\mathbb{C}}^{\oplus} \mathrm{Ind}_B^G {}^w \chi d\chi.$$

Sketch. We only argue this for $G = \mathrm{GL}_2$. Simply define

$$Mw(\varphi) := \left(\frac{L(0, \chi_1 \chi_2^{-1})}{\varepsilon(\chi_1 \chi_2^{-1}; \psi) L(1, \chi_1 \chi_2^{-1})} \right) \cdot \chi_1 \chi_2(-1) \cdot \frac{\chi_2 |\cdot|^{1/2}(\det g)}{L(1, \chi_2 \chi_1^{-1})} \int_{F_v^\times} \mathcal{F}_\psi \Phi_\varphi(te_2 g) \chi_1^{-1} \chi_2(t) |t| d^\times t,$$

where as usual the right-hand side is defined by meromorphic continuation from Tate's thesis. To check that Mw is a meromorphic operator, it is enough to check that, for any flat sections f_s and g_s , the function $s \mapsto \langle Mw(f_s), g_s \rangle$ is meromorphic. This is direct from the definition. ■

Notation 121. For later use, let Rw be equal to the operator

$$Rw(\varphi) := \chi_1 \chi_2(-1) \cdot \frac{\chi_2 |\cdot|^{1/2}(\det g)}{L(1, \chi_2 \chi_1^{-1})} \int_{F_v^\times} \mathcal{F}_\psi \Phi_\varphi(te_2 g) \chi_1^{-1} \chi_2(t) |t| d^\times t.$$

Roughly speaking, Rw is a "normalized" variant of Mw , and it admits holomorphic continuation.

Example 122. If χ is unramified, let $\varphi_\chi \in \text{Ind}_B^G \chi$ denote the unramified vector satisfying $\varphi_\chi(1) = 1$. Then $Rw(\varphi_\chi) \in \text{Ind}_B^{G,w} \chi$ is also K -invariant, so it is a scalar multiple of the corresponding unramified vector. A local calculation (with Mw) given by plugging in 1 shows that $Rw(\varphi_\chi) = \varphi_\chi^w$.

Remark 123. Because the Fourier transform is self-dual, one finds that $Rw \circ Rw = 1$.

Remark 124. If η is unitary, then $R_\eta w$ is unitary, which we see by using $Rw \circ Rw = 1$ and the adjoint property for Mw .

Remark 125. Consider the case where $\chi_1 \chi_2^{-1} = |\cdot|$ so that $\text{Ind}_B^G w \chi$ admits a one-dimensional subspace. Set $\rho := \chi_1 |\cdot|^{-1/2}$; then the one-dimensional subspace is $g \mapsto \chi_v(\det g)$. Now, $Rw: \text{Ind}_B^G \chi \rightarrow \text{Ind}_B^{G,w} \chi$ is nonzero, so its image must be a quotient of $\text{Ind}_B^G \chi$ and a subrepresentation of $\text{Ind}_B^{G,w} \chi$. Passing to the Jacquet module, we see that Rw must be a scalar times

$$\text{Ind}_B^G \chi \twoheadrightarrow \rho \circ \det \subseteq \text{Ind}_B^{G,w} \chi.$$

In particular, $Rw(\varphi) = c(\varphi)(\rho \circ \det)$ for some scalar $c(\varphi)$, and invariance properties of c force the map $c: \text{Ind}_B^G \chi \rightarrow \mathbb{C}$ to be $\langle -, \rho \circ \det \rangle$. It follows that the matrix coefficient of Rw is

$$\langle Rw(\varphi_1), \varphi_2 \rangle \simeq \langle \varphi_1, \rho \circ \det \rangle \overline{\langle \varphi_2, \rho \circ \det \rangle},$$

where the implied constant could be computed explicitly.

4 The Cuspidal Spectrum

4.1 The Cuspidal Subspace is Closed

Definition 126. Some $\varphi \in L^2([G])$ is *cuspidal* if and only if

$$\int_{[N]} \varphi(n g) dn = 0$$

for almost every $g \in A_G \backslash G(\mathbb{A}_F)$ and each unipotent radical N of a proper parabolic subgroup P of G . We define $L_{\text{cusp}}^2([G])$ to be the subspace of cuspidal functions.

Example 127. If G admits no proper parabolic subgroups, then $L^2([G]) = L_{\text{cusp}}^2([G])$.

Definition 128 (period integral). Fix a subgroup $H \subseteq G$ and a character $\chi: A_G H(F) \backslash A_G H(\mathbb{A}_F) \rightarrow \mathbb{C}$. For continuous $\varphi: [G] \rightarrow \mathbb{C}$, we define the *period*

$$\mathcal{P}_\chi(\varphi) := \int_{A_G H(F) \backslash A_G H(\mathbb{A}_F)} \varphi(h) \chi^{-1}(h) dh$$

as long as the integral absolutely converges.

Notation 129. Fix a unipotent subgroup $H \subseteq G$ and a character $\chi: A_G H(F) \backslash A_G H(\mathbb{A}_F) \rightarrow \mathbb{C}$. We define the temporary notation

$$L^2([G])(\chi) := \{ \varphi \in L^2([G]) : \mathcal{P}_\chi(Rg(\varphi)) = 0 \text{ for almost all } g \}.$$

Example 130. Note that $L^2_{\text{cusp}}([G])$ is

$$\bigcap_{\substack{\text{parabolic } P \\ P \neq G \\ P = MN}} L^2([G])(1_N).$$

Remark 131. The condition $\mathcal{P}_\chi(Rg(\varphi)) = 0$ for almost every $g \in A_G \backslash G(\mathbb{A}_F)$ is equivalent to the condition $\mathcal{P}_\chi(Rf(\varphi)) = 0$ for all $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$. Indeed, by rearranging the order of integration,

$$\int_{A_G \backslash G(\mathbb{A}_F)} f(g) \mathcal{P}_\chi(Rg(\varphi)) dg = \mathcal{P}_\chi(Rf(\varphi)).$$

Thus, $\varphi \in L^2([G])(\chi)$ does imply $\mathcal{P}_\chi(Rf(\varphi)) = 0$. For the converse, do an approximation argument.

Definition 132 (automorphic kernel). For $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$, define $K_f \in C^\infty([G] \times [G])$ by

$$K_f(x, y) := \sum_{\gamma \in G(F)} f(x^{-1}\gamma y).$$

The sum converges because it is finite.

Remark 133. By unfolding, the integral operator given by the kernel K_f on $L^2([G])$ is exactly Rf .

Proposition 134. The cuspidal subspace $L^2_{\text{cusp}}([G]) \subseteq L^2([G])$ is closed.

Sketch. It is enough to show the operators $\varphi \mapsto \mathcal{P}_\chi(Rf(\varphi))$ are continuous, where $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$. By unfolding, we see that

$$\mathcal{P}_\chi(Rf(\varphi)) = \int_{[G]} \text{Pe}_{f, \chi \delta_G}(g) \varphi(g) dg,$$

where

$$\text{Pe}_{f, \chi}(g) := \int_{A_G H(F) \backslash A_G H(\mathbb{A}_F)} K_f(h, g) \chi^{-1}(h) dh.$$

It is thus enough to show that $\text{Pe}_{f, \chi} \in L^2([G])$, which is a direct calculation. ■

4.2 The Cuspidal Subspace is Discrete

Notation 135. Suppose G is reductive. For $\alpha \in \Delta$, let Φ_α^+ denote the collection of positive roots which have α in their support. Let $N(\Phi_\alpha^+)$ denote the product of the corresponding root spaces.

Theorem 136 (Harder). Suppose F is a function field and that G is reductive. Then any smooth $\varphi \in L^2_{\text{cusp}}([G])$ is compactly supported.

Sketch. As is typical in proofs involving Siegel sets, we have a combinatorial step, and then we apply it to the Siegel sets of interest.

1. Choose a compact open subgroup $K' \subseteq G(\mathbb{A}_F)$. Then for $t \gtrsim_{K'} 0$ and $h \in \mathfrak{S}(t)$, we claim that

$$N(\Phi_\alpha^+)(F) \cdot (N(\Phi_\alpha^+) \cap hK'h^{-1}) \stackrel{?}{=} N(\Phi_\alpha^+)(\mathbb{A}_F).$$

This is a rather intricate induction on root spaces. By using the upper central series, one can choose a filtration $\{N_i\}_i$ of N so that $N_0 = N$ and $N_1 = N(\Phi_\alpha^+)$ and $N_m = 1$ for m large enough, and each quotient N_i/N_{i+1} is isomorphic to a root space $\mathbb{G}_a$. (In fact, $[N, N_i] \subseteq N_{i+1}$.) The moral is that the filtration allows us to pass to the case where $N(\Phi_\alpha^+)$ is isomorphic to $\mathbb{G}_a$, and then we conclude by using the compactness of $F \backslash \mathbb{A}_F$.

2. We now conduct the proof. By formalism involving Siegel sets, it is enough to show that $\varphi(g) = 0$ for $g \in \mathfrak{S}(t)$, where $t \gtrsim_\varphi 0$. By smoothness, φ is invariant under some compact open $K' \subseteq G(\mathbb{A}_F)$. Then choose t as in the previous step. Now, for any $h \in \mathfrak{S}(t)$ and $g \in A_G G(\mathbb{A}_F)^1 \backslash G(\mathbb{A}_F)$, we see that the function $N(\Phi_\alpha^+)(\mathbb{A}_F) \rightarrow \mathbb{C}$ defined by

$$x \mapsto \varphi(xhg_i)$$

is invariant under $N(\Phi_\alpha^+)(\mathbb{A}_F)$ (by the lemma), so they must vanish uniformly because φ is cuspidal. ■

Corollary 137. Suppose F is a function field and that G is reductive. For $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$, the operator Rf on $L^2_{\text{cusp}}([G])$ is trace class.

Sketch. By the Dixmier–Malliavin theorem, it is enough to show that Rf is Hilbert–Schmidt.

Now, the idea is that the continuity of K_f means that Rf is Hilbert–Schmidt when restricted to $L^2(\Omega)$ for any compact $\Omega \subseteq [G]$. To be precise, choose compact open $K \subseteq G(\mathbb{A}_F)$ for which $f \in C_c^\infty(G(\mathbb{A}_F))^K$. Then Rf projects $L^2_{\text{cusp}}([G])$ onto the K -invariants, so it is enough to check that Rf is Hilbert–Schmidt on a basis of $L^2_{\text{cusp}}([G])^K$. By Harder’s theorem, all such functions are compactly supported in some $\Omega \subseteq G(\mathbb{A}_F)$, so we are done because $Rf|_{L^2(\Omega)}$ is Hilbert–Schmidt. ■

Theorem 138. Suppose F is a number field and that G is reductive. Choose $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$ and $B > 0$ and $t > 0$. For any $\varphi \in L^2_{\text{cusp}}([G])$ and $x \in \mathfrak{S}(t)$, we have

$$|Rf(\varphi)(x)| \lesssim \rho(s_x)^{-B} \|\varphi\|_2.$$

Sketch. This is a hard theorem! By multiplying the inequality by itself, we may replace ρ by α , where $\alpha \in \Delta$. By restricting scalars, we may as well take $F = \mathbb{Q}$, but we will avoid using this hypothesis.

1. We separate frequencies. By unfolding,

$$Rf(\varphi)(x) = \int_{A_G N(F) \backslash G(\mathbb{A}_F)} \left(\sum_{\gamma \in N(F)} f(x^{-1}\gamma y) \right) \varphi(y) dy.$$

Fix a nontrivial character $\psi: [\mathbb{G}_a] \rightarrow S^1$. By Poisson summation,

$$Rf(\varphi)(x) = \int_{A_G N(F) \backslash G(\mathbb{A}_F)} \underbrace{\left(\sum_{\eta \in \mathfrak{n}(\mathbb{Q})} \int_{\mathfrak{n}(\mathbb{A}_F)} f(x^{-1} \exp(n)y) \psi(\langle \eta, n \rangle) dn \right)}_{\widehat{f}_\eta(x^{-1} \exp(n)y) :=} \varphi(y) dy.$$

2. We now apply the cuspidal hypothesis to get rid of some of the frequencies: we claim that we can remove the $\eta \in \mathfrak{n}(\mathbb{Q})$ for which $\langle \eta, \mathfrak{n}(\Phi_\alpha^+) \rangle = 0$. Well, for such η , an application of the Baker–Campbell–Hausdorff formula shows that $y \mapsto \hat{f}_\eta(x^{-1} \exp(n)y)$ is left-invariant by $N(\Phi_\alpha^+)(\mathbb{A}_F)$. Such a variable change is then able to produce an integral over $[N(\Phi_\alpha^+)]$ inside the integral for $Rf(\varphi)(x)$ at these frequencies.
3. There is some very delicate combinatorics of Siegel sets that we will mostly avoid, but let's give the output. We may assume that f factors as $f_\infty \otimes f^\infty \in C_c^\infty(A_G \backslash G(F_\infty)) \otimes C_c^\infty(G(\mathbb{A}_F^\infty))$. By keeping track of supports (and then the Fourier transforms), we find that

$$\int_{\mathfrak{n}(\mathbb{A}_F^\infty)} f^\infty(x^{\infty, -1} \exp(n)y_\infty) \psi(\langle \eta, n \rangle) dn \lesssim 1_{\Omega_n^\infty}(\eta)$$

for some compact Ω_n^∞ . More combinatorics of the supports shows that

$$\int_{\mathfrak{n}(F_\infty)} f_\infty(x_\infty^{-1} \exp(n)y_\infty) \psi(\langle \eta, n \rangle) dn \lesssim \delta_P(s_x) \|\text{Ad}^\vee(s_x^{-1}) \eta\|^{-B}.$$

4. It remains to lower-bound $\|\text{Ad}^\vee(s_x^{-1}) \eta\|$ when $\eta \in (\mathbb{A}_F) \cap \Omega_n^\infty$; in other words, η lives in a lattice. By considering the diagonal action of the adjoint action on the root spaces, we find $\|\text{Ad}^\vee(s_x^{-1}) \eta\| \gtrsim_t \|\eta\|_n$; alternatively, computing this action on a single root $\lambda \in \Phi_\alpha^+$ gives $\|\text{Ad}^\vee(s_x^{-1}) \eta\| \gtrsim \lambda(s_x) \|\eta\|_\lambda$, and this norm is constantly lower-bounded by being in a lattice. Combining the two bounds gives

$$\int_{\mathfrak{n}(F_\infty)} f_\infty(x_\infty^{-1} \exp(n)y_\infty) \psi(\langle \eta, n \rangle) dn \lesssim \delta_P(s_x) \alpha(s_x)^{-B} \|\eta\|^{-B} 1_{\Omega_n^\infty}.$$

Summing and integrating bounds $|Rf(\varphi)(x)|$ in terms of an L^1 norm of φ , and we conclude by Hölder's inequality. ■

Corollary 139. Suppose F is a number field and that G is reductive. For $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$, the operator Rf on $L_{\text{cusp}}^2([G])$ is trace class.

Sketch. By the Dixmier–Malliavin theorem, it is enough to show that Rf is Hilbert–Schmidt. The theorem implies that $\|Rf(\varphi)\|_\infty \lesssim_f \|\varphi\|_2$. The idea is to realize Rf as the integral operator of a kernel $K \in L^2([G] \times [G])$. For each $x \in [G]$, we may use the Riesz representation theorem to define $K_x \in L^2([G])$ so that

$$Rf(\varphi)(x) = \int_{[G]} K_x(y) \varphi(y) dy.$$

Now, define $K: [G] \times [G] \rightarrow \mathbb{C}$ (almost everywhere) by $K(x, y) := K_x(y)$. By construction Rf is given by the kernel K , so it remains to check $\|K\|_2$. Well, the bound from the theorem gives $\|K_x\|_2 \lesssim_f 1$, so one finds that $\|K\|_2$ is finite. ■

5 Decomposition of the Spectrum

5.1 Eisenstein Series

Definition 140 (discrete spectrum). Let G be a reductive group. Let $L_{\text{disc}}^2([G])$ be the largest closed subspace which decomposes into a Hilbert direct sum of irreducible subrepresentations.

Remark 141. Note that $L_{\text{disc}}^2([G])$ exists by Zorn's lemma. It contains $L_{\text{cusp}}^2([G])$.

Notation 142. Let G be a reductive group with parabolic subgroup $P = MN$. Choose a discrete subrepresentation $\sigma \subseteq L^2_{\text{disc}}([M])$. Following Arthur, we define the vector space $\text{Ind}_P^G \sigma$ to consist of the measurable functions $N(\mathbb{A}_F)A_M M(\mathbb{A}_F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ for which

- the map $m \mapsto \varphi(mx)$ is in σ for all x , and
- the norm

$$\|\varphi\|^2 := \int_K \int_{A_M M(F) \backslash M(\mathbb{A}_F)} |\varphi(mk)|^2 dm dk$$

is finite.

Given $\lambda \in \mathfrak{a}_{M\mathbb{C}}^*$, we define the representation $I(\sigma, \lambda)$ on the space $\text{Ind}_P^G \sigma$ as

$$R_\lambda g(\varphi)(x) := \varphi(xg) \exp(\langle H_P(xg) - H_P(x), \lambda + \rho_P \rangle).$$

Remark 143. One could alternatively view $I(\sigma, \lambda)$ as $\text{Ind}_P^G (\sigma \otimes e^{\langle H_P(-), \lambda \rangle})$.

Remark 144. A direct calculation shows that $I(\sigma, \lambda)$ is unitary if $\lambda \in i\mathfrak{a}_M^*$.

Notation 145. Let G be a reductive group with parabolic subgroup $P = MN$. Choose a discrete subrepresentation $\sigma \subseteq L^2_{\text{disc}}([M])$. Define $\text{Ind}_P^G(\sigma)^\circ$ to consist of the smooth K -finite vectors of $\text{Ind}_P^G(\sigma)$.

Definition 146 (Eisenstein series). Let G be a reductive group with parabolic subgroup $P = MN$. Choose an irreducible $\sigma \subseteq L^2_{\text{disc}}([M])$. For $\varphi \in \sigma$, we define the *Eisenstein series*

$$E(x, \varphi, \lambda) := \sum_{\gamma \in P(F) \backslash G(F)} \varphi(\gamma x) e^{\langle H_P(\gamma x), \lambda + \rho_P \rangle}.$$

Proposition 147. Let G be a reductive group with parabolic subgroup $P = MN$. Choose an irreducible $\sigma \subseteq L^2_{\text{cusp}}([M])$. Then $E(x, \varphi, \lambda)$ converges absolutely and uniformly on compacts of $G(\mathbb{A}_F)$ if $\text{Re } \lambda$ lives in a suitable Weyl cone.

Sketch. Because σ is cuspidal, φ is rapidly decreasing and in particular bounded. Thus, it remains to bound

$$\sum_{\gamma \in P(F) \backslash G(F)} |m_P(\gamma x)^{\lambda + \rho_P}|.$$

Let $U \subseteq G(\mathbb{A}_F)^1$ be a small open relatively compact neighborhood of 1. Then uniform continuity allows us to replace each summand with an integral over U , so we want to bound

$$\int_{P(F) \backslash G(F) \backslash U} |m_P(yx)^{\lambda + \rho_P}| dy$$

after unfolding. Because $P \backslash G$ is projective, $P(\mathbb{A}_F) \backslash G(\mathbb{A}_F)$ is compact, so we need to bound the integral over $P(F) \backslash P(\mathbb{A}_F)$. Unipotent radicals are compact, so we may even bound the integral over $M(F) \backslash M(\mathbb{A}_F)$. By passing to a Siegel set, we may go to the integral over $A_M(F) \backslash A_M(\mathbb{A}_F)$. Thus, we are basically integrating over a split torus, which will be bounded for suitable exponents λ . ■

Remark 148. A more careful analysis of how x varies over a Siegel set can show that $E(x, \varphi, \lambda)$ has moderate growth for these λ .

Theorem 149. Let G be a reductive group with parabolic subgroup $P = MN$. Choose an irreducible $\sigma \subseteq L_{\text{cusp}}^2([M])$. For $\varphi \in \sigma$, the Eisenstein series $E(x, \varphi, \lambda)$ admits meromorphic continuation to all λ .

Sketch. We only argue this for $G = \text{GL}_2$, and we will do it later. ■

5.2 Intertwining Operators

Notation 150. Let $M' \subseteq M$ be Levi subgroups of G . Then define $\mathfrak{a}_M^* := X^*(M)_{\mathbb{R}}$ and $\mathfrak{a}_M$ as its dual, and define $\mathfrak{a}_{M'}$ similarly. Then $\mathfrak{a}_{M'}^M$ is the kernel of the projection $\mathfrak{a}_{M'} \rightarrow \mathfrak{a}_M$ (induced by the dual of restriction).

Definition 151. Let G be a reductive group, and let M and M' be standard Levi subgroups of G .

- Define $W^{M, M'}$ to consist of the $w \in W$ for which $w\Phi^+(M) \subseteq \Phi^+(G)$ and $w^{-1}\Phi^+(M') \subseteq \Phi^+(G)$.
- Define $W(M, M')$ to consist of the $w \in W^{M, M'}$ for which $wMw^{-1} \subseteq M'$.

Remark 152. It turns out that W^{M_I, M_J} uniquely represents the coset space $W_J \backslash W / W_I$ by elements of minimal length.

Remark 153. The Bruhat decomposition upgrades to

$$G = \bigsqcup_{w \in W_I \backslash W / W_J} P_I w P_J.$$

Remark 154. An explicit calculation on the Lie algebra (with root spaces) shows that $w^{-1}P_J w \cap M_I$ is a standard parabolic subgroup of M_I with Levi decomposition

$$w^{-1}P_J w \cap M_I = (w^{-1}M_J w \cap M_I) (w^{-1}N_J w \cap M_I).$$

Definition 155 (associate). Two parabolic subgroups P and P' of G are associate if and only if their Levi subgroups are conjugate by G .

Notation 156. Let M and M' be standard Levi subgroups of a reductive group G . Let $W(\mathfrak{a}_M, \mathfrak{a}_{M'})$ consist of those $w \in W$ inducing an isomorphism $\mathfrak{a}_M \rightarrow \mathfrak{a}_{M'}$.

Remark 157. Let $P_0 = M_0 N_0$ be a minimal parabolic. If P and P' are standard associate with Levi subgroups M and M' , one can show that there is $w \in W$ such that the automorphism $w: \mathfrak{a}_{M_0} \rightarrow \mathfrak{a}_{M_0}$ which restricts to an isomorphism $\mathfrak{a}_M \rightarrow \mathfrak{a}_{M'}$. (Such a w also conjugates the Levi subgroups to each other.)

Definition 158. Let P and P' be associate standard parabolic subgroups of a reductive group G with $P = MN$ and $P' = M'N'$, and choose $w \in W(\mathfrak{a}_M, \mathfrak{a}_{M'})$. For irreducible $\sigma \subseteq L^2_{\text{disc}}([M])$ and $\lambda \in \mathfrak{a}_{M, \mathbb{C}}$, define $M_\lambda w: I(\sigma, \lambda) \rightarrow I(\sigma, w\lambda)$ by having $M_\lambda w(\varphi)(x)$ equal

$$\int_{(N'(\mathbb{A}_F) \cap wN(\mathbb{A}_F)w^{-1}) \backslash N'(\mathbb{A}_F)} \varphi(w^{-1}nx) \exp(\langle H_P(w^{-1}nx), \lambda + \rho_P \rangle + \langle H_{P'}(x), -w\lambda + \rho_{P'} \rangle) dn,$$

provided that the integral converges.

Remark 159. By folding the integral into a sum over F -points, one can turn this into an integral over an Eisenstein series over a compact group. Thus, for $\text{Re } \lambda$ in a suitable cone, the integral converges absolutely.

Notation 160. Fix a reductive group G . Let $\mathcal{P}$ be an association class of standard parabolic subgroups of G . Then let $\mathcal{H}_{\mathcal{P}}$ denote the Hilbert space of tuples of measurable functions $\{F_P\}_{P \in \mathcal{P}}$ such that

- $F_P: i\mathfrak{a}_{M\mathbb{R}}^{G*} \rightarrow \text{Ind}_P^G L^2_{\text{disc}}([M])$ satisfies $F_{P'}(w\lambda) = M_\lambda w F_P(\lambda)$ if $w \in W(\mathfrak{a}_m, \mathfrak{a}_{M'})$, and
- The norm

$$\|F\|^2 := \sum_{P \in \mathcal{P}} n_P^{-1} \int_{i\mathfrak{a}_{M\mathbb{R}}^{G*}} \|F_P(\lambda)\|^2 d\lambda$$

is finite, where $n_P := \sum_{P' \in \mathcal{P}} \#W(\mathfrak{a}_M, \mathfrak{a}_{M'})$.

Notation 161. Fix a reductive group G , and let $\mathcal{P}$ be an association class of standard parabolic subgroups. Define $E_{\mathcal{P}}: \mathcal{H}_{\mathcal{P}} \rightarrow L^2([G])$ by

$$E_{\mathcal{P}}(F)(x) := \sum_{P \in \mathcal{P}} n_P^{-1} \int_{i\mathfrak{a}_{M\mathbb{R}}^{G*}} E(x, F_P(\lambda), \lambda) d\lambda.$$

The image of $E_{\mathcal{P}}$ is denoted $L_{\mathcal{P}}([G])$.

Theorem 162 (Langlands). Fix a reductive group G . Then

$$L^2([G]) = \bigoplus_{\mathcal{P}} L_{\mathcal{P}}^2([G]),$$

where the sum is over all association classes (of standard parabolics).

Sketch. This theorem is extremely hard. We will say nothing about it. ■

5.3 Preliminaries for GL_2

Notation 163. Throughout, we fix a unitary central character $\omega: Z(\mathbb{A}_F) \rightarrow \mathbb{C}^\times$, which we assume has $\omega|_{F_\infty^+} = 1$.

Definition 164 (P -series). For $f: N(\mathbb{A}_F)P(F)(Z(\mathbb{A}_F), \omega) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$, define the P -series

$$F_f(x) := \sum_{\gamma \in P(F) \backslash G(F)} f(\gamma x).$$

Remark 165. If f is compactly supported modulo $N(\mathbb{A}_F)Z(\mathbb{A}_F)P(F)$, then the series is finite.

Lemma 166. Suppose $\psi \in C_c^\infty(G(F)\backslash G(\mathbb{A}_F), \omega)$, then

$$\langle \psi, F_f \rangle = \int_{P(F)Z(\mathbb{A}_F)N(\mathbb{A}_F)\backslash G(\mathbb{A}_F)} \psi_N(x) \overline{f(x)} dx.$$

Sketch. This is a direct calculation. The idea is to fold $\langle \psi, F_f \rangle$ into

$$\int_{\overline{G}(F)\backslash \overline{G}(\mathbb{A}_F)} \left(\sum_{\gamma \in P(F)\backslash G(F)} \psi(\gamma g) \overline{f(\gamma g)} \right) dg = \int_{P(F)Z(\mathbb{A}_F)\backslash G(\mathbb{A}_F)} \psi(x) \overline{f(x)} dx,$$

from which the lemma follows. ■

Remark 167. It follows that the P -series are dense in $L^2_{\text{cusp}}(G(F)\backslash G(\mathbb{A}_F), \omega)^\perp$.

Notation 168. For $f: N(\mathbb{A}_F)P(F)(Z(\mathbb{A}_F), \omega)\backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$, define

$$\tilde{f}(g) := \int_{N(\mathbb{A}_F)} f(wng) dn.$$

We may also denote this integral operator by Mw .

Remark 169. The Bruhat decomposition (and unfolding) implies $F_{f,N}(g) = f(g) + \tilde{f}(g)$, so

$$\langle F_{f_1}, F_{f_2} \rangle = \int_{P(F)Z(\mathbb{A}_F)N(\mathbb{A}_F)\backslash G(\mathbb{A}_F)} f_1(x) \overline{f_2(x)} dx + \int_{P(F)Z(\mathbb{A}_F)N(\mathbb{A}_F)\backslash G(\mathbb{A}_F)} \tilde{f}_1(x) \overline{\tilde{f}_2(x)} dx.$$

Notation 170. For $f: N(\mathbb{A}_F)P(F)(Z(\mathbb{A}_F), \omega)\backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$, we define the *Mellin transform*

$$\hat{f}(g; s) := \int_{F_\infty^+} f \left(\begin{bmatrix} t & \\ & 1 \end{bmatrix} g \right) |t|^{-s-1/2} d^\times t.$$

Remark 171. Note $\hat{f}(-; \lambda)$ descends to $N(\mathbb{A}_F)T(F)(Z(\mathbb{A}_F), \omega)\backslash G(\mathbb{A}_F)$. Thus, it makes sense to define the inner product

$$\langle \hat{f}_1(s), \hat{f}_2(-\bar{s}) \rangle := \int_K \int_{F^\times \backslash \mathbb{A}_F^1} \hat{f}_1(\text{diag}(a, 1)k; s) \overline{\hat{f}_2(\text{diag}(a, 1)k; -\bar{s})} da dk.$$

Proposition 172. For $f_1, f_2: N(\mathbb{A}_F)P(F)(Z(\mathbb{A}_F), \omega)\backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ and $x > 1/2$, one has

$$\langle F_{f_1}, F_{f_2} \rangle = \frac{1}{2\pi i} \int_{\text{Re } \lambda = x} \langle \hat{f}_1(\lambda), \hat{f}_2(-\bar{\lambda}) \rangle d\lambda + \frac{1}{2\pi i} \int_{\text{Re } \lambda = x} \langle M_\lambda w(\hat{f}_1)(\lambda), \hat{f}_2(\bar{\lambda}) \rangle d\lambda.$$

Sketch. Applying the Iwasawa decomposition and then Mellin inversion yields

$$\int_{P(F)Z(\mathbb{A}_F)N(\mathbb{A}_F)\backslash G(\mathbb{A}_F)} f_1(x) \overline{f_2(x)} dx = \frac{1}{2\pi i} \int_{\operatorname{Re} \lambda = x} \langle \widehat{f_1}(\lambda), \widehat{f_2}(-\bar{\lambda}) \rangle d\lambda.$$

One has a similar expression available with f_1 replaced with $\tilde{f}_1$. Rearranging w inside the integral completes the proof. ■

5.4 Decomposition for GL_2

Notation 173. Let $\mathcal{H}$ be the Hilbert space bundle over $\mathbb{C}$ whose fiber $\mathcal{H}_\lambda$ over $\lambda \in \mathbb{C}$ consists of functions $f: N(\mathbb{A}_F)P(F)(Z(\mathbb{A}_F), \omega) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ satisfying

$$f(\operatorname{diag}(u, v)g) = |u/v|^{\lambda+1/2} \varphi(g),$$

and satisfying a suitable finiteness condition.

Remark 174. One can further diagonalize $\mathcal{H}_\lambda$ into irreducible representations, but we will frequently not bother to do so.

Notation 175. Let $\chi: [T] \rightarrow \mathbb{C}^\times$ be a character. Define $R_\chi w: \operatorname{Ind}_P^G \chi \rightarrow \operatorname{Ind}_P^G w \chi$ to be the (restricted) tensor product of the local operators.

Remark 176. This restricted tensor product is well-defined because it sends the distinguished unramified vector to the distinguished unramified vector when everything is unramified. This is a bounded operator because $R_{\chi_v} w$ has norm one for cofinitely many v .

Remark 177. By expanding out the local calculations, we find $Rw \circ Rw = 1$, and $R_\chi w$ is unitary whenever χ is unitary.

Theorem 178. The global operator

$$Mw: \int_{X^*(T)_\mathbb{C}}^{\oplus} \operatorname{Ind}_P^G \chi d\chi \rightarrow \int_{X^*(T)_\mathbb{C}}^{\oplus} \operatorname{Ind}_P^G w \chi d\chi$$

admits meromorphic continuation, with at worst simple poles located along $s = 1/2$, where $\chi_1 \chi_2^{-1} = |\cdot|^s$.

Sketch. Simply define

$$M_\chi w := \frac{L(0, \mu\nu^{-1})}{L(1, \mu\nu^{-1}) \varepsilon(0, \mu\nu^{-1})} \cdot R_\chi w,$$

which is equal to the restricted tensor product of the $M_{\chi_v} w$ s. ■

Remark 179. Using the global functional equation, one sees that $Mw \circ Mw = 1$. Thus, $M_\chi w$ is unitary when χ is unitary.

Remark 180. The residue of the pole of $M_\chi w$ at $s = 1/2$ can be understood in terms of the inner products

$$\text{Res}_\chi \langle Mw(\widehat{f}_1), \widehat{f}_2 \rangle \simeq \langle R_\chi w(\widehat{f}_1)(\chi), \widehat{f}_2(\chi) \rangle = \langle \widehat{f}_1, \chi \circ \det \rangle \langle \widehat{f}_2, \chi \circ \det \rangle.$$

The last equality follows from the local calculation. The relevant constant can be made explicit.

Theorem 181. One has that $L^2(\text{GL}_2(F) \backslash \text{GL}_2(\mathbb{A}_F), \omega)$ is isomorphic to

$$L^2_{\text{cusp}}(\text{GL}_2(F) \backslash \text{GL}_2(\mathbb{A}_F), \omega) \oplus \bigoplus_{\chi_1 \chi_2 = \omega} \widehat{\int_{i\mathbb{R}/\langle \pm \rangle}^\oplus \text{Ind}_B^G(\chi_1 |\cdot|^\lambda, \chi_2 |\cdot|^{-\lambda}) d\lambda} \oplus \bigoplus_{\chi^2 = \omega} (\chi \circ \det).$$

Sketch. The second term is the diagonalization of the (global sections of the) bundle $\mathcal{H}$. Anyway, by moving the vertical line of integration, we find that

$$\begin{aligned} \langle F_{f_1}, F_{f_2} \rangle &= \frac{1}{2\pi i} \int_{\text{Re } s=0} \langle \widehat{f}_1(s) + M_s w \widehat{f}_1(-s), \widehat{f}_2(-\bar{s}) \rangle ds \\ &\quad + c \sum_{\chi^2 = \omega} \langle \widehat{f}_1(1/2), \chi \circ \det \rangle \overline{\langle \widehat{f}_2(1/2), \chi \circ \det \rangle}, \end{aligned}$$

where the second sum accumulates the relevant poles. Note that $\langle \widehat{f}_1(1/2), \chi \circ \det \rangle$ unfolds into $\langle F_1, \chi \circ \det \rangle$. We are now in a position to define our isometry.

- Define a to send $C_c^\infty(N(\mathbb{A}_F)P(F)Z(\mathbb{A}_F) \backslash G(\mathbb{A}_F), \omega)$ to the direct integral by sending f to the section $a_f(iy) := \frac{1}{2}(\widehat{f}(iy) + M_{iy} w(\widehat{f}(-iy)))$, which we note satisfies $M_{iy} w(a_f(iy)) = a_f(-iy)$.
- If $\chi^2 = \omega$, there is a map $C_c^\infty(N(\mathbb{A}_F)P(F)Z(\mathbb{A}_F) \backslash G(\mathbb{A}_F), \omega) \rightarrow \chi \circ \omega$ given by $f \mapsto \langle \widehat{f}(1/2), \chi \circ \det \rangle$.

The calculation above shows that $F_\bullet: C_c^\infty(N(\mathbb{A}_F)P(F)Z(\mathbb{A}_F) \backslash G(\mathbb{A}_F), \omega) \rightarrow L^2_{\text{cusp}}(\text{GL}_2(F) \backslash \text{GL}_2(\mathbb{A}_F), \omega)^\perp$ and the map

$$(a, \widehat{f}(1/2)) : C_c^\infty(N(\mathbb{A}_F)P(F)Z(\mathbb{A}_F) \backslash G(\mathbb{A}_F), \omega) \rightarrow \bigoplus_{\chi_1 \chi_2 = \omega} \widehat{\int_{i\mathbb{R}/\langle \pm \rangle}^\oplus \text{Ind}_B^G(\chi_1 |\cdot|^\lambda, \chi_2 |\cdot|^{-\lambda}) d\lambda} \oplus \bigoplus_{\chi^2 = \omega} (\chi \circ \det)$$

produces the same norm, so their completions are isomorphic. The theorem follows. $\blacksquare$

6 The Spectral Expansion

6.1 The Cuspidal Spectrum

Lemma 182. Choose $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$, and define f^* by $f^*(g) := \overline{f(g^{-1})}$. Then $(Rf)^* = R(f^*)$ as operators on $L^2([G])$.

Sketch. Expand

$$\langle Rf(\varphi), \psi \rangle = \int_{[G]} \int_{A_G \backslash G(\mathbb{A}_F)} f(g) \varphi(xg) \overline{\psi(x)} dg dx.$$

Sending $x \mapsto xg^{-1}$ and then $g \mapsto g^{-1}$ gives the result. $\blacksquare$

Proposition 183. Choose a subrepresentation $V \subseteq L^2_{\text{disc}}([G])$. For any $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$, the operator $Rf|_V$ admits a smooth kernel.

Sketch. Fix a basis $\{\varphi_i\}_i$ of V . Then define

$$K_{f,V}(x, y) := \sum_i Rf(\varphi_i)(x) \overline{\varphi_i(y)}.$$

This is the desired kernel once we view $Rf|_V$ as the composite $L^2([G]) \xrightarrow{f} V \subseteq L^2([G])$. Smoothness is checked by the Dixmier–Malliavin theorem, which lets us reduce to the case where $f = f_1 * f_2 * f_3$, so $K_{f,V}$ can be viewed as a translate of the group action by f_1 and f_2 . ■

Definition 184 (relative trace). Let $H \subseteq G \times G$ be a subgroup for which $H^\circ = H_u \times H_r$; such an H is called *almost reductive*. Define $A_{G,H} := H(\mathbb{A}_F) \cap (A_G \times A_G)$. For a character $\chi: A_{G,H}H(F) \backslash H(\mathbb{A}_F) \rightarrow \mathbb{C}^\times$ and cuspidal $\pi \subseteq L^2_{\text{cusp}}(A_G \backslash G(\mathbb{A}_F))$ and $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$, the *relative trace* is

$$\text{rtr}_\chi \pi(f) := \int_{A_{G,H}H(F) \backslash H(\mathbb{A}_F)} K_{\pi(f)}(h) \chi^{-1}(h) dh.$$

Remark 185. This integral absolutely converges, but it is a little tricky to show. In short, we may pass to the connected H case by a finite-index argument, and then we may pass to the reductive H case because $[H_u]$ is compact. Via Siegel sets, one can basically pass to the case where H is a split torus, and then the result follows upon expanding $K_{\pi(f)}(x, y) = \sum_i Rf(\varphi_i)(x) \overline{\varphi_i(y)}$ because functions in V are rapidly decreasing.

Remark 186. Intuitively, rtr_χ is a period against χ . Accordingly, one can show that $\pi \otimes \pi^\vee$ is χ -distinguished if and only if the functional $\text{rtr}_\chi \pi: C_c^\infty(A_G \backslash G(\mathbb{A}_F)) \rightarrow \mathbb{C}$. The main difficulty lies in upgrading a generic function to a smooth compactly supported one.

Theorem 187 (Cuspidal expansion). Fix $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$. Let $H \subseteq G \times G$ be almost reductive, and choose a character $\chi: A_{G,H}H(F) \backslash H(\mathbb{A}_F) \rightarrow \mathbb{C}^\times$. Then for $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$,

$$\int_{A_{G,H}H(F) \backslash H(\mathbb{A}_F)} K_{f,\text{cusp}}(h) \chi^{-1}(h) dh = \sum_{\text{cuspidal } \pi} \text{rtr}_\chi \pi(f).$$

Sketch. Once granted absolute convergence, this follows because a basis expansion implies

$$K_{f,\text{cusp}} = \sum_{\pi} K_{\pi(f)}$$

almost everywhere. As usual, by the Dixmier–Malliavin theorem, we may reduce to the case where $f = f_1 * f_2 * f_3$. Then one finds that

$$K_{\pi(f)} = (\pi(f_2) \times \pi(\overline{f_1^*})) K_{\pi(f_3)}.$$

We thus find that $K_{\pi(f)}$ is rapidly decreasing, which is enough to give the required absolute convergence. ■

Remark 188. Using the full decomposition of $L^2([G])$, one has a spectral expansion

$$K_f(x, y) = \sum_{P=MN} n_P^{-1} \sum_{\sigma \subseteq L^2_{\text{disc}}([M])} \int_{i\mathfrak{a}_M^{G*}} \left(\sum_{\varphi \in \mathcal{B}(\sigma)} E(x, I(\sigma, \lambda)(f)(\varphi), \lambda) \overline{E(y, \varphi, \lambda)} \right),$$

where the sum is over all standard parabolic subgroups, and $\mathcal{B}(\sigma)$ is a chosen basis of σ . Indeed, this is essentially the content of the decomposition of $L^2([G])$ (and its relationship with Eisenstein series).

6.2 The Residual Spectrum for GL_2

Definition 189 (truncation). Choose $T > 0$. For $\varphi: [G] \rightarrow \mathbb{C}$, we define the *truncation*

$$\wedge^T \varphi(g) := \varphi(g) - \sum_{\gamma \in P(F) \backslash G(F)} \varphi_N(\gamma g) 1_{[T, \infty)}(\log H(\gamma g)),$$

where we recall $H(n \text{diag}(a, b)k) = |a/b|$. An argument on Siegel sets shows that the sum is locally finite.

Example 190. If φ is cuspidal, the constant term vanishes, so $\wedge^T \varphi = \varphi$.

Remark 191. For $g \in G(\mathbb{A}_F)$ satisfying $H(g) > 1$, one can show that any $\gamma \in G(F)$ with $H(\gamma g) > 1$ must have $\gamma \in P(F)$. One can use this to show that

$$\wedge^T \varphi(g) = \varphi(g) - \varphi_N(g) 1_{[T, \infty)}(\log H(g))$$

if $H(g) > 1$.

Remark 192. Intuitively, the truncation kills constant terms, so it has rapid decrease (e.g., if φ is smooth).

Remark 193. For $T > 0$, the operator $\wedge^T$ is self-adjoint: indeed, a direct expansion and unfolding gives

$$\langle \wedge^T \varphi_1, \varphi_2 \rangle = \langle \varphi_1, \varphi_2 \rangle - \int_{P(F)N(\mathbb{A}_F)Z(\mathbb{A}_F) \backslash G(\mathbb{A}_F)} \varphi_{1N}(g) \overline{\varphi_{2N}(g)} 1_{[T, \infty)}(\log H(g)) dg.$$

Remark 194. One can show that $\wedge^T \wedge^T \varphi = \wedge^T \varphi$ by direct calculation. The point is that $(\wedge^T \varphi)_N(g) = 0$ if $\log H(g) > T$.

Notation 195. Let $K \in L^2([G] \times [G])$ be a function. Then we define $\wedge_2^T K(x, y)$ to denote truncation in the second coordinate.

Example 196. For $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$, we see that $\wedge_2^T K_{f, \text{cusp}}(x, y) = K_{f, \text{cusp}}(x, y)$ by direct expansion of the kernel on a basis.

Lemma 197. For $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$, we have

$$\lim_{T \rightarrow \infty} \int_{[G]} \wedge_2^T K_{f, \text{res}}(x, x) dx = \text{tr} (Rf; L^2_{\text{res}}([G])).$$

Sketch. Note that Rf is compact on the residual spectrum, so the trace equals

$$\int_{[\overline{G}]} K_{f,\text{res}}(x, x) dx = \text{tr} (Rf; L_{\text{res}}^2([G])).$$

Thus, the difficulty lies in removing the truncation. By expanding on a basis, we see that

$$K_{f,\text{res}}(x, y) \simeq \sum_{\chi^2=\omega} \left(\chi(\det x) \overline{\chi(\det y)} \int_{\overline{G}(\mathbb{A}_F)} f(g) \chi(\det g) dg \right).$$

Truncating in y has the effect of multiplying the χ summand by $1 - \sum_{\gamma} 1_{[T,\infty)}(\exp H(\gamma y))$. Thus, one needs to bound

$$\int_{\overline{G}(\mathbb{A}_F)} \left(\sum_{\gamma \in P(F) \backslash G(F)} 1_{[T,\infty)}(\exp H(\gamma x)) \right) dx.$$

As $T \rightarrow \infty$, one shows that this vanishes by unfolding the integral and integrating via the Iwasawa decomposition. ■

6.3 The Continuous Spectrum for GL_2

Notation 198. Following earlier discussion, given a holomorphic section φ of $\mathcal{H}$, we define the *Eisenstein series* as

$$E(x, \varphi, \lambda) := \sum_{\gamma \in P(F) \backslash G(F)} \varphi_{\lambda}(\gamma x).$$

This is an example of a P -series.

Remark 199. By Mellin inversion, for $f \in C_c^\infty(N(\mathbb{A}_F)(Z(\mathbb{A}_F), \omega)T(F) \backslash G(\mathbb{A}_F))$, we see $F_f(g)$ is

$$\sum_{\gamma \in P(F) \backslash G(F)} f(\gamma g) = \frac{1}{2\pi i} \sum_{\gamma \in P(F) \backslash G(F)} \int_{\text{Re } \lambda = x} \widehat{f}_s(\gamma g) d\lambda = \frac{1}{2\pi i} \int_{\text{Re } \lambda = x} E(g, \widehat{f}, \lambda) d\lambda.$$

Thus, we will be able to use Eisenstein series to discuss the continuous spectrum.

Theorem 200. Let φ be a flat section of $\mathcal{H}$. Then $E(x, \varphi, \lambda)$ admits meromorphic continuation to all $\mathbb{C}$, and its poles agree with the poles of $E_N(x, \varphi, \lambda)$.

Sketch. Note that $E_N(g, \varphi, \lambda) = \varphi_{\lambda}(g) + M_{\lambda} w(\varphi_{\lambda})(g)$, so it admits meromorphic continuation. For the continuation, we use Whittaker expansion. To this end, we fix a nontrivial character $\psi: [N] \rightarrow \mathbb{C}^\times$, and we would like to bound the Fourier coefficients

$$\int_{[N]} E(n g, \varphi, \lambda) \psi_{\alpha}(n)^{-1} dn$$

for $\alpha \in F$. If $\alpha = 0$, we get the constant term. For $\alpha \in F^\times$, unfolding with the Bruhat decomposition yields that the Fourier coefficient is essentially

$$\int_{N(\mathbb{A}_F)} \varphi_{\lambda}(w n \text{diag}(\alpha, 1) g) \psi(-n) dn.$$

Now, by density, we may as well assume that φ_{λ} is a pure tensor and Schwartz–Bruhat, so the integral factors. At finite places v , it vanishes if $|\alpha|_v$ is large enough to keep φ_{λ} oscillating enough, so the Fourier coefficients are supported in a fractional ideal of F . At the infinite places, we are basically looking at a genuine Fourier coefficient, which is still Schwartz. Convergence follows. ■

Corollary 201. Let φ be a flat section of $\mathcal{H}$. Then $E(x, \varphi, \lambda)$ satisfies the functional equation

$$E(x, Mw(\varphi), g) = E(x, \varphi, g).$$

Sketch. Both sides have the same constant term, so the difference is cuspidal. But the difference is also a P -series, so it is orthogonal to $L^2_{\text{cusp}}([G])$. ■

Notation 202. Temporarily, define S to be the projection from $L^2([G])$ to the global sections of $\mathcal{H}$, so S factors through the projection to the continuous spectrum.

Lemma 203. Fix $f \in C_c^\infty((Z(\mathbb{A}_F), \omega)T(F)N(\mathbb{A}_F)\backslash G(\mathbb{A}_F))$ and let φ be a flat section of $\mathcal{H}$. Then

$$\langle \varphi, SF_f(iy) \rangle = \frac{1}{2} \int_{G(F)Z(\mathbb{A}_F)\backslash G(\mathbb{A}_F)} E(h_{iy}, g) \overline{F_f(g)} dg.$$

Sketch. Choose f_1 and f_2 , and define $F_i := F_{f_i}$ for brevity. The idea is to compute $\langle F_1, F_2 \rangle$ in two ways. On one hand, earlier we computed

$$\langle F_1, F_2 \rangle = \frac{1}{\pi} \int_{\mathbb{R}} \langle a_{f_1}(iy), a_{f_2}(iy) \rangle dy + c \sum_{\chi^2=\omega} \langle F_1, \chi \circ \det \rangle \overline{\langle F_2, \chi \circ \det \rangle}.$$

On the other hand, we can expand F_1 into an integral of Eisenstein series, which gives

$$\langle F_1, F_2 \rangle = \frac{1}{2\pi i} \int_{\text{Re } s=x} \left(\int_{\overline{G}(F)\backslash \overline{G}(\mathbb{A}_F)} E(g, \widehat{f}_1, s) \overline{F_2(g)} dg \right) ds$$

for $x > 1/2$. Moving the line of integration to $x = 0$, we pick up residues from poles, which are the poles of the constant term E_N , which all come from the intertwining operator. Thus, we get

$$\langle F_1, F_2 \rangle = \frac{1}{2\pi} \int_{\mathbb{R}} \left(\int_{\overline{G}(F)\backslash \overline{G}(\mathbb{A}_F)} E(g, a_{f_1}, iy) \overline{F_2(g)} dg \right) dy + c \sum_{\chi^2=\omega} \langle F_1, \chi \circ \det \rangle \overline{\langle F_2, \chi \circ \det \rangle},$$

where we have implicitly used the functional equation to replace $\widehat{f}_1$ with a_{f_1} . Comparing the two expressions and letting f_1 vary completes the proof. ■

Proposition 204. Fix $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$. Let $\{\varphi_\alpha\}$ be a basis of $\mathcal{H}_0$, which we extend to flat sections. Then

$$K_{f, \text{cont}}(g, h) = \frac{1}{4\pi} \sum_{\alpha, \beta} \int_{\mathbb{R}} \langle \pi_{iy}(f) \varphi_\beta, \varphi_\alpha \rangle E(h, \varphi_\alpha, iy) \overline{E(g, \varphi_\beta, iy)} dy.$$

Sketch. Note that S^*S just projects $L^2([G])$ onto the continuous spectrum. Thus, after some rearrangement, it is enough to show that

$$\langle S^*S(Rf)S^*SF_1, F_2 \rangle \stackrel{?}{=} \iint_{[G]^2} F_1(g) \overline{F_2(h)} \left(\frac{1}{4\pi} \sum_{\alpha} \int_{\mathbb{R}} E(h, \varphi_\alpha, iy) \overline{E(g, f * \varphi_\alpha, iy)} dy \right) dg dh.$$

Well, rewrite the left-hand side as $\langle f * SF_1, SF_2 \rangle$, and then expand out the inner product in terms of a basis of $\mathcal{H}_0$ and apply the lemma. ■

Lemma 205. Fix flat sections φ_1 and φ_2 of $\mathcal{H}$. Then

$$\langle \wedge^T E(\varphi_1, s), \wedge^T E(\varphi_2, -\bar{s}) \rangle = 2T \langle \varphi_1, \varphi_2 \rangle + \langle M_{-s} M'_s \varphi_1, \varphi_2 \rangle + \frac{1}{2s} \left(\langle \varphi_1, M_{-\bar{s}} \varphi_2 \rangle e^{2sT} - \langle M_s \varphi_1, \varphi_2 \rangle e^{-2sT} \right).$$

Sketch. Work with generic $s_1, s_2 \in \mathbb{C}$ for a moment. Because we are taking an inner product with a P -series, we see

$$\langle E(\varphi_1, s_1), \wedge^T E(\varphi_2, \bar{s}_2) \rangle$$

can be expanded into an integral against the constant term of $E(\varphi_1, s_1)$. This constant term is basically $\varphi_1 + Mw(\varphi_1)$, so we have brought everything down to integrals of φ_1 and φ_2 . After using the Iwasawa decomposition to simplify these integrals, we are left with

$$\begin{aligned} & \left(\langle \varphi_1, \varphi_2 \rangle e^{(s_1+s_2)T} - \langle M_{s_1} w(\varphi_1), M_{\bar{s}_2} w(\varphi_2) \rangle e^{-(s_1+s_2)T} \right) \frac{1}{s_1 + s_2} \\ & + \left(\langle \varphi_1, M_{\bar{s}_2} w(\varphi_2) \rangle e^{(s_1-s_2)T} - \langle M_{s_1} w(\varphi_1), \varphi_2 \rangle e^{-(s_1-s_2)T} \right) \frac{1}{s_1 - s_2}. \end{aligned}$$

Sending $s_2 \rightarrow -s_1$ and applying a version of L'Hôpital's rule completes the proof. ■

Proposition 206. Fix $f \in C_c^\infty(A_G \backslash G(\mathbb{A}_F))$. Then

$$\begin{aligned} \int_{[\overline{G}]} \wedge_2^T K_{f,\text{cont}}(g, g) dg &= T \text{vol}(F^\times \backslash \mathbb{A}_F^1) \sum_{\gamma \in \overline{M}(F)} \int_{M(\mathbb{A}_F) \backslash G(\mathbb{A}_F)} f(g^{-1} \gamma g) dg \\ &\quad - \frac{1}{4\pi} \int_{\mathbb{R}} \text{tr} M_{-iy} M'_{iy} \pi_{iy}(f) dy + \frac{1}{4} \text{tr} M_0 \pi_0(f) + o_T(1). \end{aligned}$$

Sketch. This is a very long calculation. Plugging into the expansion of $K_{f,\text{cont}}$, we see that

$$\begin{aligned} \int_{[\overline{G}]} \wedge_2^T K_{f,\text{cont}}(g, g) dg &= \frac{T}{2\pi} \sum_{\alpha, \beta} \int_{\mathbb{R}} \langle \pi_{iy}(f) \varphi_\beta, \varphi_\alpha \rangle \langle \varphi_\alpha, \varphi_\beta \rangle dy \\ &\quad - \frac{1}{4\pi} \sum_{\alpha, \beta} \int_{\mathbb{R}} \langle \pi_{iy}(f) \varphi_\beta, \varphi_\alpha \rangle \langle M_{-iy} M'_{iy} \varphi_\alpha, \varphi_\beta \rangle dy \\ &\quad + \frac{1}{4\pi} \sum_{\alpha, \beta} \int_{\mathbb{R}} \langle \pi_{iy}(f) \varphi_\beta, \varphi_\alpha \rangle \left(\langle \varphi_\alpha, M_{iy} \varphi_\beta \rangle \frac{e^{2Tiy}}{2iy} - \langle M_{iy} \varphi_\alpha, \varphi_\beta \rangle \frac{e^{-2Tiy}}{2iy} \right) dy. \end{aligned}$$

The terms are ordered to align with the proposition.

- The first sum rearranges to

$$\frac{T}{2\pi} \int_{\mathbb{R}} \text{tr} \pi_{iy}(f) dy.$$

One can directly check that the action of $\pi_{iy}(f)$ on $\mathcal{H}_{iy}$ is given by the kernel

$$\int_{N(\mathbb{A}_F)} \int_{\mathbb{R}^+} \sum_{\alpha \in F^\times} f(x^{-1} n \text{diag}(\alpha t, 1) k) |t|^{-\frac{1}{2}+iy} dn d^\times t$$

as a function of x and the pair $(a, k) \in F^\times \backslash \mathbb{A}_F^1 \times K$. (We need to use the Iwasawa decomposition.) Integrating the kernel along the diagonal gives the first term of the term of the proposition.

- Rearrangemnt shows that the second sum corresponds to the second term in the proposition.

- The third term rearranges to

$$\frac{1}{4\pi} \int_{\mathbb{R}} \operatorname{tr} M_{-iy} \pi_{iy}(f) \cdot \frac{e^{2Ty} - e^{-2Ty}}{2iy} dy + \frac{1}{4\pi} \int_{\mathbb{R}} \frac{\operatorname{tr} M_{-iy} \pi_{iy}(f) - \operatorname{tr} M_{iy} \pi_{iy}(f)}{2iy} \cdot e^{-2Ty} dy.$$

The right term is a Fourier coefficient, so it goes to zero as $T \rightarrow \infty$. The left term is computed using differentiation under the integral sign (by differentiating with respect to T). ■